

Formalizing building-up construction of self-dual codes through isotropic lines in Lean

Jae-Hyun Baek

Department of Artificial Intelligence and
Institute for Mathematical and Data Sciences
Sogang University, Seoul, Korea

Jon-Lark Kim

Department of Mathematics and
Institute for Mathematical and Data Sciences
Sogang University, Seoul, Korea

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Abstract

We study building-up constructions of Euclidean self-dual codes through isotropic lines and formalize their algebraic structure in Lean 4. For $q \equiv 1 \pmod{4}$, we prove that every self-dual code over $\mathbb{F}_q$ has a repeated rank- r boxed generator after a permutation of two-coordinate blocks. The terminal $r \times r$ matrix is nonsingular, and explicit Gram equations characterize self-duality. Simultaneous removal of pivot rows and coordinate blocks preserves this form; conversely, an explicit extension can be iterated while keeping the terminal matrix fixed. We determine precisely when such an extension is a Kim–Lee step over the same parent, and identify the complementary direct-sum case. In the binary case, we give an independent formal proof of the Chinburg–Zhang rank-one boxed representation and compare its reduction with Kim’s construction. The arithmetic pairing is transported through a specified orthonormal-basis isometry; the distinguished Euclidean coordinate plane has Gram matrix I_2 , not the alternating hyperbolic Gram matrix. The arithmetic realization theorem remains a cited input. Applications include optimal self-dual $[6, 3, 4]$ and $[8, 4, 4]$ codes over $\text{GF}(5)$, together with MDS self-dual codes over $\text{GF}(13)$. In particular, a known pure double-circulant $[14, 7, 8]$ code is put into an explicit universal rank-one box; its two-coordinate reduction gives a self-dual $[12, 6, 6]$ parent with a unique projective correction capable of restoring distance 8. Reproducible finite-field computations accompany these examples; they are separate from the kernel-checked algebraic proofs.

1 Introduction

Coding theory studies reliable communication, following Shannon’s foundational work [20]. Among its classical examples, the extended binary Hamming code of length 8, the extended binary Golay code of length 24, and the extended ternary Golay code of length 12 are self-dual, that is, $C = C^\perp$. Thus the classification or construction of self-dual codes has been one of the most active research problems [18]. Self-dual codes have wide connections with several mathematical areas including t -design theory [1], group theory [6], number theory [2], [6], [21]. Furthermore, they have applications in quantum error-correcting codes [4].

Kim [9] introduced “building-up construction” which proved that all binary self-dual codes can be constructed in this way. This construction starts from a binary self-dual code of length $2n$ and then adds a vector of length $2n$ of odd weight to construct a binary self-dual code of length $2n+2$. Kim–Lee [12], [11] generalized this method to construct self-dual codes over finite fields $\mathbb{F}_q$, where q is even or $q \equiv 1 \pmod{4}$. In practice, this is a principal method for constructing examples over these finite fields. However, the formula for the new generator rows is often presented as a random vector whose inner product with itself is -1 . One of the central points of the present paper is that a correct row vector is naturally organized by an isotropic line in a hyperbolic plane, and that the forward construction admits a precise row-wise and family-wise characterization in those terms.

Kreck and Puppe [13] realized binary self-dual codes using the cohomology of three-manifolds. Chinburg and Zhang [5] obtained an arithmetic counterpart from the restriction map for μ_2 , the sheaf of square roots of unity, on a ring of S -integers. Its image is self-annihilating for a bilinear pairing assembled from local Hilbert symbols. When the local pairings admit orthonormal bases, the image becomes a Euclidean binary self-dual code in those coordinates. Their arithmetic existence theorem, quoted as Theorem 3.7, realizes every binary self-dual code of length at least four up to permutation equivalence. We use this cited arithmetic input only through the restriction image, its pairing, and a chosen form isometry; these objects are specified before Theorem 3.9. Our algebraic reduction and reconstruction do not require a formalization of arithmetic duality.

In this paper, we focus on $q \equiv 1 \pmod{4}$. Over $\mathbb{F}_q$, self-dual codes are maximal totally isotropic subspaces for the standard Euclidean bilinear form on $\mathbb{F}_q^{2n}$. In this split case the elementary identity $c^2 = -1$ is the basic algebraic input behind both the classical building-up construction and the structural description of the ambient quadratic space: it produces an isotropic line in $\mathbb{F}_q^2$, identifies the Euclidean plane with a hyperbolic plane, and makes the norm form $x^2 + y^2$ split.

We establish that each lower generator row of the building-up admits the isotropic line decomposition $r_i = -y_i \cdot (1, c, 0, \dots, 0) + (0, 0, g_i)$, where $(1, c)$ spans an isotropic line in the hyperbolic plane. This decomposition provides a precise geometric interpretation of the correction terms in the building-up construction. We also give an explicit representation by the quadratic form $x^2 + y^2$: once $c^2 = -1$ and $2 \neq 0$, every $a \in \mathbb{F}_q$ can be written explicitly as $a = x^2 + y^2$, which in turn yields extension vectors of prescribed norm.

The two arithmetic inputs must be distinguished. In the odd-characteristic q -ary part, $q \equiv 1 \pmod{4}$ makes -1 a square in $\mathbb{F}_q^\times$, so the Euclidean plane is hyperbolic. In the binary construction of Chinburg–Zhang [5, Theorem 1.2 and Corollary 1.4], the local cup-product pairing admits an orthonormal basis under the stated local hypotheses; for a non-complex place this is equivalent to -1 being a nonsquare in K_v . After the resulting coordinate isometry is fixed, the arithmetic image becomes a standard Euclidean self-dual binary code and its boxed matrix can be compared with Kim’s construction.

Kim’s binary building-up theorem and converse, the Kim–Lee extensions, and the Chinburg–Zhang arithmetic realization and binary boxed representation are established results. We distinguish them from the structural results proved here: the universal rank- r representation over split odd fields, its recursive restrictions and extensions, and its fixed-parent correspondence with Kim–Lee steps. The binary representation is given an independent algebraic proof suitable for formalization. The Lean development verifies these algebraic arguments and their interfaces; it does not formalize the cited arithmetic realization theorem.

Our main contributions are as follows.

- (1) In the binary case, we separate the cohomological pairing from the standard Euclidean dot product by an explicit form isometry. In Euclidean coordinates the first coordinate plane is non-alternating with Gram matrix I_2 ; its diagonal vector $(1, 1)$ is isotropic, and orthogonality forces the Kim correction formula. We prove intrinsically that every binary Euclidean self-dual code is permutation equivalent to the literal rank-one boxed form, by two-coordinate reduction and row normalization of the Kim extension (Theorem 3.8). Deleting the first two coordinates and the top row recovers a shorter Euclidean self-dual code, and exact rebuilding recovers the boxed code (Theorem 3.9).
- (2) In the q -ary case, we refine the classical building-up construction into an exact self-duality criterion and prove a universal rank- r boxed normal form. Its lower-right block is specified by a nonsingular $r \times r$ matrix D , subject to the displayed Gram relations. Simultaneous pivot-row and block-column deletion preserves the form and self-duality (Theorem 3.6); its rank-one specialization with zero pivot diagonal recovers the explicit split boxed construction (Corollary 3.10). Lemma 3.1 constructs repeatable extensions with the same D , and Lemmas 3.2–3.3 identify their fixed-parent correspondence with Kim–Lee steps, including the direct-sum exception.
- (3) As concrete applications, we exhibit the largest certified repeated boxed presentations over $\text{GF}(2)$, $\text{GF}(5)$, and $\text{GF}(13)$, ending at self-dual codes with parameters $[24, 12, 8]$, $[24, 12, 9]$, and $[22, 11, 10]$, respectively. For a known pure double-circulant $[14, 7, 8]$ code over $\text{GF}(13)$, we also give an exact universal rank-one generator, its $[12, 6, 6]$ two-coordinate parent, and the unique projective correction that can restore distance 8 over this parent (Section 4).
- (4) We formalize the algebraic results in Lean 4, including the self-duality criteria, binary two-coordinate reduction, the universal normal forms, and the conditional coefficient

relations. The formal statements distinguish coordinate permutations from isometries between different bilinear forms.

Formalization in Lean

The accompanying Lean 4 development formalizes the finite-dimensional linear and bilinear algebra used in the paper. It includes the self-dual/Lagrangian characterization, the systematic generator criterion, the building-up constructions, both universal normal forms, the explicit fixed-parent correspondence between repeated boxes and Kim–Lee steps, and the conditional coefficient relations of Theorem 3.11. In the binary case, the proof of the universal normal form proceeds by two-coordinate reduction and induction; it does not assume a classification theorem. The quoted results of Kim, Kim–Lee, and Chinburg–Zhang remain cited mathematical results and are not introduced as Lean axioms. The finite-field searches and minimum-distance computations in the applications are separate computations, not formalized proofs.

The Lean listings state the corresponding definitions and theorems; the complete proofs and verification records are available at

`LeGenAI/intersection-coding-theory-cohomology`.

The development uses Lean `v4.29.0-rc6` and Mathlib. All 123 independently compared declarations are accepted by Lean’s default kernel. Their only foundational axioms are `propext`, `Quot.sound`, and `Classical.choice`; the proofs use neither `sorry` nor `native_decide`. All seventeen suites were replayed together in a fresh Linux project snapshot on August 27, 2026. All 123 declarations passed statement and dependency comparison, the permitted-axiom audit, and default-kernel replay. The recorded inputs were unchanged throughout the run. The numerical example certificates and the external minimum-distance computations also passed, but are not included in the 123-goal count.

The paper is organized as follows. Section 2 fixes notation and the split quadratic background. Section 3 collects the building-up tools and presents the universal rank- r boxed form, its binary rank-one refinement, and the normalized rank-one forms over odd fields. Section 4 gives the certified applications over $\text{GF}(2)$, $\text{GF}(5)$, and $\text{GF}(13)$. Section 5 closes with the comparison viewpoint and the formalization perspective.

2 Preliminaries

We refer to [8] for basic concepts and facts and to [18] for self-dual codes.

Throughout the paper, $\mathbb{F}_q$ denotes a finite field. We equip $\mathbb{F}_q^n$ with the Euclidean bilinear

form

$$\langle u, v \rangle := \sum_{i=1}^n u_i v_i.$$

Here “Euclidean” refers to the coding-theoretic dot product, not to positive definiteness. We use $\mathbb{F}_q$ for a general finite field and $\text{GF}(5)$, $\text{GF}(13)$ for the concrete fields in the examples. For a linear code, the minimum distance equals the minimum Hamming weight of its nonzero words. For a linear code $C \subseteq \mathbb{F}_q^n$, we write

$$C^\perp := \{v \in \mathbb{F}_q^n : \langle v, c \rangle = 0 \text{ for all } c \in C\}.$$

Definition 2.1 (self-dual code) A linear code $C \subseteq \mathbb{F}_q^n$ is *self-dual* if $C = C^\perp$ [17].

Definition 2.2 (Lagrangian subspace) Let V be a finite-dimensional vector space over $\mathbb{F}_q$ equipped with a nondegenerate symmetric bilinear form. A subspace $L \subseteq V$ is called *Lagrangian* [15, 19] if $L = L^\perp$.

For a nondegenerate form, $\dim L^\perp = \dim V - \dim L$, so $L = L^\perp$ if and only if L is totally isotropic and $2 \dim L = \dim V$. In particular, V is even-dimensional whenever it admits a Lagrangian subspace. Applied to the standard Euclidean form, this shows that a code $C \subseteq \mathbb{F}_q^n$ is self-dual if and only if it is totally isotropic and $2 \dim C = n$. Thus a self-dual code is precisely a Lagrangian subspace of $(\mathbb{F}_q^n, \langle \cdot, \cdot \rangle)$, and its length is necessarily even.

Listing 1: Lean for Definitions 2.1 and 2.2.

```
def paperSelfDualCode (C : Submodule K (Fin n → K)) : Prop :=
  C = (dotBilin (K := K) (n := n)).orthogonal C

def paperLagrangianSubspace (L : Submodule K (Fin n → K)) : Prop :=
  L = (dotBilin (K := K) (n := n)).orthogonal L

theorem paper_self_dual_iff_lagrangian :
  paperSelfDualCode (K := K) C ↔ paperLagrangianSubspace (K := K) C

theorem paper_dotBilin_nondegenerate :
  (dotBilin (K := K) (n := n)).Nondegenerate

theorem paperLagrangianSubspace_iff_totallyIsotropic_and_finrank_half :
  paperLagrangianSubspace (K := K) L ↔
    IsTotallyIsotropic (dotBilin (K := K) (n := n)) L ∧
    2 * Module.finrank K L = Module.finrank K (Fin n → K)

theorem paperSelfDualCode_iff_totallyIsotropic_and_finrank_half :
  paperSelfDualCode (K := K) C ↔
    IsTotallyIsotropic (dotBilin (K := K) (n := n)) C ∧
    2 * Module.finrank K C = Module.finrank K (Fin n → K)

theorem paperLagrangianSubspace_even_length
```

```

(hL : paperLagrangianSubspace (K := K) L) : Even n

theorem paperSelfDualCode_even_length
(hC : paperSelfDualCode (K := K) C) : Even n

```

For the odd-characteristic q -ary construction below, we further assume $q \equiv 1 \pmod{4}$.

Proposition 2.1 (Systematic form criterion) *Let $C \subseteq \mathbb{F}_q^{2k}$ be generated by a matrix $G = [I_k \mid A]$ with $A \in M_k(\mathbb{F}_q)$. Then C is self-dual if and only if*

$$AA^T = -I_k.$$

Proof. The rows of G span a k -dimensional code. Hence self-duality is equivalent to pairwise orthogonality of the rows. The Gram matrix GG^T of the rows is

$$GG^T = I_k + AA^T = O,$$

so the orthogonality condition is exactly $AA^T = -I_k$. □

Listing 2: Lean for Proposition 2.1.

```

def systematicRows (A : Matrix (Fin k) (Fin k) K) :
  Fin k → Fin (k + k) → K :=
  fun i => Fin.append (Pi.single i 1) (A i)

theorem systematicRows_linearIndependent (A : Matrix (Fin k) (Fin k) K) :
  LinearIndependent K (systematicRows A)

theorem paper_systematic_form_criterion_exact
(A : Matrix (Fin k) (Fin k) K) :
  paperSelfDualCode (rowSpace (systematicRows A)) ↔
  A * A.transpose = -(1 : Matrix (Fin k) (Fin k) K)

```

Definition 2.3 (Permutation Equivalence) Two codes $C_1, C_2 \subseteq \mathbb{F}_q^n$ are *permutation equivalent* if there is a permutation $\sigma \in S_n$ such that $C_2 = P_\sigma C_1$, where $(P_\sigma u)_j = u_{\sigma(j)}$.

Coordinate permutations are Euclidean isometries:

$$\langle P_\sigma u, P_\sigma v \rangle = \langle u, v \rangle.$$

Consequently, permutation equivalence preserves Euclidean self-duality. A general monomial transformation additionally permits independent nonzero coordinate scalings; it need not preserve the Euclidean form or self-duality and is not included in our use of “equivalence” below. Over $\mathbb{F}_2$, permutation and monomial equivalence coincide because $\mathbb{F}_2^\times = \{1\}$.

Listing 3: Lean for Definition 2.3.

```

def CodeEquiv (G1 : Fin 1m → Fin n → K)
  (G2 : Fin 2m → Fin n → K) : Prop :=
  ∃ σ : Equiv.Perm (Fin n),
    rowSpace (permuteFamily σ G1) = rowSpace G2

theorem dot_coordinatePermuteLinearEquiv :
  dot (permuteVec σ u) (permuteVec σ v) = dot u v

theorem codeEquiv_preserves_paperSelfDualCode
  (hG1G2 : CodeEquiv G1 G2)
  (hG1 : paperSelfDualCode (K := K) (rowSpace G1)) :
  paperSelfDualCode (K := K) (rowSpace G2)

```

From this point onward, when discussing the split q -ary construction we assume that q is odd and satisfies $q \equiv 1 \pmod{4}$. The condition $q \equiv 1 \pmod{4}$ is equivalent to -1 being a square in $\mathbb{F}_q^\times$. We fix once and for all an element

$$c \in \mathbb{F}_q^\times \quad \text{with} \quad c^2 = -1.$$

This choice is used throughout the building-up formulas.

Definition 2.4 (Hyperbolic Plane) A *hyperbolic plane* [19] over $\mathbb{F}_q$ is a 2-dimensional symmetric bilinear space with basis $\{e_1, e_2\}$ satisfying

$$\langle e_1, e_1 \rangle = \langle e_2, e_2 \rangle = 0, \quad \langle e_1, e_2 \rangle = 1.$$

Such a basis $\{e_1, e_2\}$ will be called a *hyperbolic pair*. Any 1-dimensional totally isotropic subspace is called an *isotropic line*.

In even dimension and odd characteristic, a nondegenerate symmetric bilinear space is called *split* if it is an orthogonal direct sum of hyperbolic planes. Thus the splitting condition concerns the bilinear space, not the underlying finite field.

Proposition 2.2 *Let q be odd and let $c \in \mathbb{F}_q$ satisfy $c^2 = -1$. Then:*

- (i) *The element c has multiplicative order 4.*
- (ii) *The Euclidean plane $(\mathbb{F}_q^2, \langle \cdot, \cdot \rangle)$ is hyperbolic.*
- (iii) *For every $k \geq 1$, the matrix equation $AA^T = -I_k$ has the explicit solution $A = cI_k$.*
- (iv) *The quadratic form $x^2 + y^2$ splits as*

$$x^2 + y^2 = (x - cy)(x + cy),$$

hence it represents every element of $\mathbb{F}_q$.

Proof. Since q is odd, $-1 \neq 1$. From $c^2 = -1$ we obtain

$$c^4 = (c^2)^2 = (-1)^2 = 1, \quad c^2 = -1 \neq 1.$$

Thus the order of c divides 4, but does not divide 2, and therefore $\text{ord}(c) = 4$. Put

$$e_1 = (1, c), \quad e_2 = (2^{-1}, -2^{-1}c).$$

Then

$$\langle e_1, e_1 \rangle = 1 + c^2 = 0, \quad \langle e_2, e_2 \rangle = 2^{-2}(1 + c^2) = 0,$$

whereas

$$\langle e_1, e_2 \rangle = 2^{-1}(1 - c^2) = 2^{-1}(1 - (-1)) = 1.$$

If $\alpha e_1 + \beta e_2 = 0$, taking inner products first with e_2 and then with e_1 gives

$$\alpha = \langle \alpha e_1 + \beta e_2, e_2 \rangle = 0, \quad \beta = \langle \alpha e_1 + \beta e_2, e_1 \rangle = 0.$$

Hence e_1, e_2 are linearly independent and consequently form a hyperbolic basis of the 2-dimensional Euclidean plane.

For $A = cI_k$, symmetry of I_k and $c^2 = -1$ give

$$AA^T = (cI_k)(cI_k)^T = c^2 I_k = -I_k.$$

Finally,

$$(x - cy)(x + cy) = x^2 - c^2 y^2 = x^2 + y^2.$$

For an arbitrary $a \in \mathbb{F}_q$, take

$$x = \frac{1+a}{2}, \quad y = \frac{1-a}{2}c.$$

Since $c^2 = -1$, we have

$$cy = -\frac{1-a}{2}, \quad x - cy = 1, \quad x + cy = a,$$

and hence $x^2 + y^2 = (x - cy)(x + cy) = a$. □

Proposition 2.3 *Let q be odd and let $c \in \mathbb{F}_q$ satisfy $c^2 = -1$. Then the vectors*

$$e_1 = (1, c), \quad e_2 = (2^{-1}, -2^{-1}c)$$

form a hyperbolic basis of $\mathbb{F}_q^2$. Equivalently, the line spanned by $(1, c)$ is isotropic and the Euclidean plane is split.

Proof. The proof of Proposition 2.2(ii) computes $\langle e_i, e_i \rangle = 0$, $\langle e_1, e_2 \rangle = 1$, and establishes linear independence of these two vectors. These are exactly the required basis and Gram conditions. □

Listing 4: Lean for Proposition 2.2 and Proposition 2.3.

```

theorem paper_split_consequences_exact
  [Fact ((2 : K) ≠ 0)] (c : K) (hc : c ^ 2 = (-1 : K)) :
  orderOf c = 4 ∧
  (paperHyperbolicPair (splitE1 c) (splitE2 c) ∧
   LinearIndependent K ![splitE1 c, splitE2 c]) ∧
  ∀( k : Nat,
    (c • (1 : Matrix (Fin k) (Fin k) K)) *
      (c • (1 : Matrix (Fin k) (Fin k) K)).transpose =
        -(1 : Matrix (Fin k) (Fin k) K)) ∧
  ∀( a : K, ∃ x y : K,
    x - c * y = 1 ∧ x + c * y = a ∧ x ^ 2 + y ^ 2 = a)

theorem norm_form_witness_factors
  [Fact ((2 : K) ≠ 0)] (a c : K) (hc : c ^ 2 = (-1 : K)) :
  let x := (1 + a) / 2
  let y := ((1 - a) / 2) * c
  x - c * y = 1 ∧ x + c * y = a ∧ x ^ 2 + y ^ 2 = a

theorem paper_euclidean_plane_hyperbolic_basis_exact
  [Fact ((2 : K) ≠ 0)] (c : K) (hc : c ^ 2 = (-1 : K)) :
  paperHyperbolicPair (splitE1 c) (splitE2 c) ∧
  LinearIndependent K ![splitE1 c, splitE2 c]

```

A change to hyperbolic coordinates is distinct from permutation equivalence. Suppose $2 \neq 0$ and $c^2 = -1$, and write $v = (v_1, v_2, v')$. The invertible linear map

$$A_c(v) = \left(\frac{v_1 - cv_2}{2}, \frac{v_1 + cv_2}{2}, v' \right)$$

satisfies $B_{\text{sp}}(A_c u, A_c v) = \langle u, v \rangle$, where

$$B_{\text{sp}}(u, v) = 2(u_1 v_2 + u_2 v_1) + u' \cdot v'.$$

Consequently, $C = C^{\perp_E}$ if and only if $A_c C = (A_c C)^{\perp_{B_{\text{sp}}}}$. The map A_c does not preserve the Euclidean form: $A_c(1, c, 0) = (1, 0, 0)$, whose self-pairing is 1, whereas $(1, c, 0)$ has self-pairing 0. Thus A_c is not a Euclidean isometry and cannot replace a coordinate permutation in our normal-form theorems.

3 Building-Up and Boxed Normal Forms

We first collect the building-up constructions and the self-duality criterion used below. We then organize the boxed forms by their normalization: the universal rank- r pattern in Section 3.2, the binary rank-one refinement in Section 3.3, and the normalized rank-one construction over odd fields in Section 3.4. Their computational realizations are collected separately in Section 4.

3.1 Building-up constructions

In 2001, Kim [9] proposed a building-up construction of binary self-dual codes in order to construct many self-dual codes of larger lengths from a self-dual code of a given length. An explicit form of the building-up construction is given below.

Theorem 3.1 (*[9, Theorem 1]*) *Let G_0 be a generator matrix of a binary self-dual code C_0 of length $2n$, whose rows are $g_1, \dots, g_n$. Let $\mathbf{x}$ be a binary vector in $\mathbb{F}_2^{2n}$ satisfying $\mathbf{x} \cdot \mathbf{x} = 1$. Suppose that $y_i = \mathbf{x} \cdot g_i$ for $1 \leq i \leq n$. Then the following matrix*

$$G = \left[\begin{array}{cc|c} 1 & 0 & \mathbf{x} \\ \hline y_1 & y_1 & g_1 \\ \vdots & \vdots & \vdots \\ y_n & y_n & g_n \end{array} \right]$$

generates a binary self-dual code C of length $2n + 2$.

Proof. Write

$$r_0 = (1, 0, \mathbf{x}), \quad r_i = (y_i, y_i, g_i) \quad (1 \leq i \leq n).$$

Because C_0 is self-dual, its generator rows $g_1, \dots, g_n$ are linearly independent and satisfy $g_i \cdot g_j = 0$ for all i, j . In characteristic two,

$$\begin{aligned} r_0 \cdot r_0 &= 1 + \mathbf{x} \cdot \mathbf{x} = 1 + 1 = 0, \\ r_0 \cdot r_i &= y_i + \mathbf{x} \cdot g_i = y_i + y_i = 0, \\ r_i \cdot r_j &= 2y_i y_j + g_i \cdot g_j = 0. \end{aligned}$$

Thus the rows of G are pairwise orthogonal.

It remains to verify their independence. Suppose

$$\lambda_0 r_0 + \sum_{i=1}^n \lambda_i r_i = 0.$$

Subtracting the second coordinate of this relation from the first gives $\lambda_0 = 0$. The tail coordinates then give

$$\sum_{i=1}^n \lambda_i g_i = 0,$$

so $\lambda_i = 0$ for every i by the independence of the rows of G_0 . Hence G has $n + 1$ independent, mutually orthogonal rows in $\mathbb{F}_2^{2n+2}$. Its row space is therefore totally isotropic of dimension $n + 1 = \frac{1}{2}(2n + 2)$, and consequently equals its orthogonal complement. $\square$

Listing 5: Exact Lean theorem for Theorem 3.1.

```

theorem paper_binary_kim_building_up_exact [CharP K 2]
  {m : Nat} {x : Fin (2 * m) → K}
  {G : Fin m → Fin (2 * m) → K}
  (hx : dot x x = (1 : K))
  (hparent : paperSelfDualCode (rowSpace G)) :
  paperSelfDualCode (rowSpace (buildRowsBin x G))

```

Then the converse of Theorem 3.1 was proved in [9] as follows.

Theorem 3.2 (*[9, Theorem 2]*) *Any binary self-dual code C of length $2n$ with minimum distance $d > 2$ is obtained from some self-dual code C_0 of length $2n - 2$ (up to permutation equivalence) by the construction in Theorem 3.1.*

For the odd-characteristic constructions in the rest of this subsection, write $K = \mathbb{F}_q$ with $q \equiv 1 \pmod{4}$.

Motivated by the building-up construction of binary self-dual codes, Kim and Lee [11] generalized it to finite fields with q elements where $q \equiv 1 \pmod{4}$ as follows.

Theorem 3.3 (*[11, Proposition 2.1]*) *Let $\mathbb{F}_q$ be a finite field with q elements. Assume that $q \equiv 1 \pmod{4}$. Let c be in $\mathbb{F}_q$ such that $c^2 = -1$. Let G_0 be a generator matrix of a Euclidean self-dual code C_0 over $\mathbb{F}_q$ of length $2n$, whose rows are $g_1, \dots, g_n$. Let $\mathbf{x}$ be a vector in $\mathbb{F}_q^{2n}$ satisfying $\mathbf{x} \cdot \mathbf{x} = -1$. Suppose that $y_i = \mathbf{x} \cdot g_i$ for $1 \leq i \leq n$. Then the following matrix*

$$G = \left[\begin{array}{cc|c} 1 & 0 & \mathbf{x} \\ \hline -y_1 & cy_1 & g_1 \\ \vdots & \vdots & \vdots \\ -y_n & cy_n & g_n \end{array} \right]$$

generates a self-dual code C over $\mathbb{F}_q$ of length $2n + 2$.

Proof. Let

$$r_0 = (1, 0, \mathbf{x}), \quad r_i = (-y_i, cy_i, g_i) \quad (1 \leq i \leq n).$$

Since $C_0 = C_0^\perp$, the rows of G_0 are linearly independent and $g_i \cdot g_j = 0$ for all i, j . Using $y_i = \mathbf{x} \cdot g_i$ and $c^2 = -1$, we obtain

$$\begin{aligned}
 r_0 \cdot r_0 &= 1 + \mathbf{x} \cdot \mathbf{x} = 1 - 1 = 0, \\
 r_0 \cdot r_i &= -y_i + \mathbf{x} \cdot g_i = 0, \\
 r_i \cdot r_j &= (-y_i)(-y_j) + (cy_i)(cy_j) + g_i \cdot g_j \\
 &= (1 + c^2)y_i y_j + g_i \cdot g_j = 0.
 \end{aligned}$$

Thus the row space of G is self-orthogonal.

For linear independence, suppose

$$\lambda_0 r_0 + \sum_{i=1}^n \lambda_i r_i = 0.$$

The second coordinate gives

$$c \sum_{i=1}^n \lambda_i y_i = 0.$$

Now $c \neq 0$, since $c^2 = -1$, and hence $\sum_i \lambda_i y_i = 0$. The first coordinate then gives $\lambda_0 - \sum_i \lambda_i y_i = \lambda_0 = 0$. Taking tail coordinates yields

$$\sum_{i=1}^n \lambda_i g_i = 0,$$

so every λ_i vanishes. Consequently G has $n + 1$ independent pairwise orthogonal rows in $\mathbb{F}_q^{2n+2}$. Its row space has dimension

$$n + 1 = \frac{2n + 2}{2},$$

and therefore equals its Euclidean orthogonal complement. □

Listing 6: Exact Lean theorem for Theorem 3.3.

```
theorem paper_qary_kim_lee_building_up_exact
  {m : Nat} {x : Fin (2 * m) -> K} {c : K}
  {G : Fin m -> Fin (2 * m) -> K}
  (hx : dot x x = (-1 : K))
  (hc : c ^ 2 = (-1 : K))
  (hparent : paperSelfDualCode (rowSpace G)) :
  paperSelfDualCode (rowSpace (buildRows x (-c) G))
```

Remark 3.1 The sign convention of Kim–Lee in the displayed matrix is $(-y_i, cy_i, g_i)$. From this point on we use the equivalent normalization with rows $(-y_i, -cy_i, g_i)$. Since $(-c)^2 = c^2 = -1$, this changes only the sign convention in the second correction coordinate.

Then the converse of the above theorem was also stated and proved in [11].

Theorem 3.4 (*[11, Proposition 2.2]*) *Let $\mathbb{F}_q$ be a finite field with q elements. Assume that $q \equiv 1 \pmod{4}$. Any Euclidean self-dual code C over $\mathbb{F}_q$ of length $2n$ with minimum distance $d > 2$ is obtained from some Euclidean self-dual code C_0 over $\mathbb{F}_q$ of length $2n - 2$ (up to permutation equivalence) by the construction method in Theorem 3.3.*

Kim and Lee define two linear codes to be permutation equivalent when a permutation of coordinates sends one code to the other, and Proposition 2.2 uses the phrase “up to

permutation equivalence” [11, pp. 81, 83–84]. The row equivalence appearing inside their proof only changes the chosen generator basis and does not enlarge the code equivalence relation to monomial equivalence.

Before imposing a recursive boxed shape, we determine precisely when the building-up matrix generates a self-dual code, without a minimum-distance assumption.

The ambient splitting used below is independent of the code and of its generator matrix. In the literal standard-coordinate decomposition

$$W = E_2 \perp W_0, \quad E_2 = K^2, \quad W_0 = K^{2n-2},$$

the plane E_2 has Gram matrix I_2 . Since the characteristic is odd, put

$$e = (1, c), \quad f = \left(\frac{1}{2}, -\frac{c}{2}\right).$$

Then

$$e \cdot e = 1 + c^2 = 0, \quad f \cdot f = \frac{1 + c^2}{4} = 0, \quad e \cdot f = \frac{1 - c^2}{2} = 1.$$

Thus (e, f) is a hyperbolic basis of E_2 , and the basis-change map gives the form isometry $E_2 \simeq H$. Consequently $W \simeq H \perp W_0$, whereas the matrix below remains written in the original standard Euclidean coordinates. This distinction is precisely Proposition 2.3; it is not inferred from the displayed generator matrix.

The next theorem combines the forward and reverse calculations. No prescribed block structure is imposed on G . The theorem determines exactly when adjoining the displayed row and two columns to G produces a generator matrix of a self-dual code.

Theorem 3.5 (Self-duality criterion for the building-up matrix) *Let q be a prime power with $q \equiv 1 \pmod{4}$, and let $K = \mathbb{F}_q$. Fix $c \in K^\times$ such that $c^2 = -1$. Let $x \in K^{2n-2}$, let $y_1, \dots, y_{n-1} \in K$, and let G be an arbitrary $(n-1) \times (2n-2)$ matrix with rows $g_1, \dots, g_{n-1}$. Form the bordered matrix*

$$G' = \left[\begin{array}{cc|c} 1 & 0 & x \\ \hline -y_1 & -cy_1 & g_1 \\ \vdots & \vdots & \vdots \\ -y_{n-1} & -cy_{n-1} & g_{n-1} \end{array} \right],$$

and put $L = \text{rowspan}(G')$ and $L_0 = \text{rowspan}(G) \subseteq K^{2n-2}$. Then

$$L = L^\perp \iff \begin{cases} x \cdot x = -1, \\ y_i = x \cdot g_i \quad (1 \leq i \leq n-1), \\ L_0 = L_0^\perp. \end{cases}$$

Whenever these equivalent conditions hold, G' is literally the normalized Kim–Lee building-up matrix obtained from G and x , not merely a generator of an equivalent code. Deleting its first row and first two columns recovers G exactly.

Proof. Suppose first that $L = L^\perp$. Since the Euclidean form on K^{2n} is nondegenerate, $\dim L = n$. The n displayed rows therefore form a basis of L . Write

$$r_0 = (1, 0, x), \quad r_i = (-y_i, -cy_i, g_i).$$

Self-orthogonality gives

$$0 = \langle r_0, r_0 \rangle = 1 + \langle x, x \rangle,$$

so $\langle x, x \rangle = -1$. Likewise,

$$0 = \langle r_0, r_i \rangle = -y_i + \langle x, g_i \rangle,$$

and hence $y_i = \langle x, g_i \rangle$ for every i . For $1 \leq i, j \leq n-1$,

$$\langle r_i, r_j \rangle = y_i y_j + c^2 y_i y_j + \langle g_i, g_j \rangle = (1 + c^2) y_i y_j + \langle g_i, g_j \rangle.$$

Since $c^2 = -1$, the first term vanishes, and thus

$$\langle r_i, r_j \rangle = \langle g_i, g_j \rangle.$$

Because L is self-orthogonal, this proves $\langle g_i, g_j \rangle = 0$ for all i, j , so L_0 is self-orthogonal.

To determine its dimension, suppose

$$\sum_{i=1}^{n-1} a_i g_i = 0.$$

The coefficient formula gives

$$\sum_{i=1}^{n-1} a_i y_i = \left\langle x, \sum_{i=1}^{n-1} a_i g_i \right\rangle = 0.$$

Consequently all three blocks of $\sum_i a_i r_i$ vanish. The displayed rows are linearly independent, so all a_i vanish. Thus the tail rows are linearly independent and

$$\dim L_0 = n - 1 = \frac{1}{2} \dim W_0.$$

The self-orthogonality of L_0 , together with this half-dimension equality and nondegeneracy of the Euclidean form on K^{2n-2} , gives $L_0 = L_0^\perp$.

Conversely, assume $x \cdot x = -1$, $y_i = x \cdot g_i$ for every i , and $L_0 = L_0^\perp$. Substitution of the coefficient identities gives the rows $(1, 0, x)$ and $(-x \cdot g_i, -c(x \cdot g_i), g_i)$. Theorem 3.3, applied with the square root $-c$, therefore shows that L is self-dual. Finally, tail projection of every row other than the first deletes the first two coordinates and returns g_i , so deletion recovers G exactly. $\square$

In the following listing, `qaryFreeCoreBoxedFamily x c Y G` denotes the displayed bordered matrix, with arbitrary G . The equivalence states the norm and coefficient conditions and the self-duality of the row space of G ; the final equalities express the building-up and deletion formulas.

Listing 7: Lean for Theorem 3.5.

```

theorem paper_qary_free_core_boxed_equivalence
  (hc : c ^ 2 = (-1 : K)) :
  paperSelfDualCode
    (rowSpace (qaryFreeCoreBoxedFamily x c Y G)) ↔
  dot x x = (-1 : K) ∧
  ∀( i, Y i = dot x (G i)) ∧
  paperSelfDualCode (rowSpace G) ∧
  qaryFreeCoreBoxedFamily x c Y G = buildRows x c G ∧
  deleteHyperbolicPairSplit
    (qaryFreeCoreBoxedFamily x c Y G) = G

```

Remark 3.2 Proposition 2.3 identifies the Euclidean plane with a hyperbolic plane independently of the generator matrix. Theorem 3.5 characterizes self-duality for arbitrary G ; it does not impose a boxed normal form on G . We turn to such normal forms next.

3.2 Universal rank- r boxed normal form

Theorem 3.5 characterizes self-duality of the displayed bordered matrix without prescribing a block structure for G . Under its equivalent conditions, the matrix is exactly the normalized Kim–Lee building-up matrix. In the minimum-distance range $d > 2$, Kim–Lee’s Proposition 2.2 further shows that every q -ary self-dual code is permutation equivalent to such a presentation. The next theorem specifies a recursive block structure for the generator matrix, designed to play for $q \equiv 1 \pmod{4}$ the same recursive role that boxed matrices play in the binary theorem of Chinburg–Zhang. In odd characteristic the terminal part need not consist of a single block column, and the pivot diagonal need not be fixed to 01. We first retain both freedoms to obtain a universal, recursively closed form, and then examine its binary refinement and normalized rank-one specialization in the following subsections.

To describe the pivot coefficients, write each two-coordinate block as $(\alpha, c\alpha + \beta)$, where $c^2 = -1$. In the first k block columns, the pivot rows are normalized so that their β -coordinates form the identity matrix. Writing P_{ij} for the corresponding α -coordinate therefore gives the block $(P_{ij}, cP_{ij} + \delta_{ij})$. Thus P is a matrix of coefficients, not a coordinate permutation; it need not be symmetric, triangular, or have zero diagonal.

Theorem 3.6 (Universal rank- r boxed normal form over $\mathbb{F}_q$) *Let $q \equiv 1 \pmod{4}$, let $K = \mathbb{F}_q$, and fix $c \in K^\times$ with $c^2 = -1$, and identify $K^{2n} = (K^2)^n$. For every Euclidean self-dual code $C \subseteq K^{2n}$, define*

$$U_c := \bigoplus_{j=1}^n K(1, c), \quad r := \dim(C \cap U_c), \quad k := n - r.$$

Then, after a permutation of the n two-coordinate blocks and elementary row operations, C has a generator matrix of the following repeated form. Its first k rows are the pivot rows,

and its last r rows form a basis of $C \cap U_c$. Write $G(c; P, H, Q, A, D)$ for this matrix and set

$$L_i = \left((H_{it}, cH_{it} + Q_{it}) \right)_{t=1}^r.$$

Then

$$G = \left[\begin{array}{cccc|c} (P_{11}, cP_{11} + 1) & P_{12}(1, c) & \cdots & P_{1k}(1, c) & L_1 \\ P_{21}(1, c) & (P_{22}, cP_{22} + 1) & \cdots & P_{2k}(1, c) & L_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ P_{k1}(1, c) & P_{k2}(1, c) & \cdots & (P_{kk}, cP_{kk} + 1) & L_k \\ \hline A_{\bullet 1}(1, c) & A_{\bullet 2}(1, c) & \cdots & A_{\bullet k}(1, c) & (D_{st}(1, c))_{r \times r} \end{array} \right], \quad (1)$$

where the upper-left block has entries $(P_{ij}, cP_{ij} + \delta_{ij})$ and $A_{\bullet j}(1, c)$ denotes the column of blocks $A_{sj}(1, c)$, $1 \leq s \leq r$. Here $P \in M_k(K)$, $H, Q \in M_{k \times r}(K)$, $A \in M_{r \times k}(K)$, and the terminal block is specified by $D \in M_r(K)$, which satisfies

$$\det D \neq 0. \quad (2)$$

The coefficients satisfy exactly

$$A + DQ^\top = 0, \quad (3)$$

$$I_k + c(P + P^\top) + c(HQ^\top + QH^\top) + QQ^\top = 0. \quad (4)$$

Since $c^{-1} = -c$, the second relation is equivalently

$$P + P^\top = c(I_k + QQ^\top) - (HQ^\top + QH^\top).$$

For fixed H, Q , it determines the symmetric part of P , while its skew-symmetric part remains arbitrary. Conversely, for every $k, r \geq 0$, every matrix $G(c; P, H, Q, A, D)$ satisfying (2)–(4) has $k + r$ linearly independent, pairwise orthogonal rows and therefore generates a Euclidean self-dual code of length $2(k + r)$.

Moreover, the form is closed under simultaneous deletion of any pivot rows and their corresponding block columns. More precisely, for any ordered subset $I \subseteq \{1, \dots, k\}$ of size l , the retained matrix is

$$G_I = G(c; P_{I,I}, H_{I,*}, Q_{I,*}, A_{*,I}, D), \quad (5)$$

and it generates a self-dual code of length $2(l + r)$, with the same terminal matrix D . At $I = \emptyset$, its row space is exactly $U_c^{(r)} := \{(z_1(1, c), \dots, z_r(1, c)) : z \in K^r\}$.

Proof. For $v = (v_1, \dots, v_n) \in (K^2)^n$, write each block uniquely as

$$v_j = (\alpha_j(v), c\alpha_j(v) + \beta_j(v)), \quad \alpha_j(v) = v_{j,1}, \quad \beta_j(v) = v_{j,2} - cv_{j,1}.$$

Let

$$\partial : C \longrightarrow K^n, \quad \partial(v) = (\beta_1(v), \dots, \beta_n(v)).$$

Since $\ker \partial = C \cap U_c$, rank-nullity and $\dim C = n$ give

$$\text{rank } \partial + \dim(C \cap U_c) = n, \quad \text{rank } \partial = k.$$

Choose a set J of k block coordinates for which the coordinate projection

$$\text{im } \partial \longrightarrow K^J$$

is an isomorphism, and permute the blocks so that $J = \{1, \dots, k\}$. Choose pivot rows $g_1, \dots, g_k \in C$ whose images under ∂ restrict to the standard basis of K^J , and choose rows $S_1, \dots, S_r$ forming a basis of $\ker \partial$. These n rows form a basis of C : applying ∂ annihilates the S_s and determines all pivot coefficients, after which independence of the S_s determines the remaining coefficients.

Reading the first coordinates and the differences β_j of these rows produces matrices P, H, Q, A, D and gives (1). It remains to prove that the terminal matrix D is nonsingular. For $S \in \ker \partial$, write $S = (a_1(1, c), \dots, a_n(1, c))$ and put $a(S) = (a_1, \dots, a_n)$. Since

$$(1, c) \cdot (s, t) = s + ct = c(t - cs),$$

orthogonality of S to every $v \in C$ gives

$$0 = S \cdot v = c a(S) \cdot \partial(v).$$

Hence

$$a(\ker \partial) \subseteq (\text{im } \partial)^\perp.$$

The map a is injective on $\ker \partial$, and both spaces have dimension $r = n - k$; therefore equality holds. If $a \in (\text{im } \partial)^\perp$ vanishes outside J , then the surjectivity of $\text{im } \partial \rightarrow K^J$ forces $a = 0$. Thus projection of $(\text{im } \partial)^\perp$ onto the complementary r coordinates is an isomorphism. In the bases $S_1, \dots, S_r$, its matrix is precisely D , proving $\det D \neq 0$.

Finally, the block identity

$$(\alpha, c\alpha + \beta) \cdot (\gamma, c\gamma + \delta) = c(\alpha\delta + \beta\gamma) + \beta\delta$$

shows that orthogonality between the rows g_i and S_s is exactly (3), while pivot-pivot orthogonality is exactly (4); products between the S_s vanish identically. This proves the existence direction. Conversely, these two relations give pairwise orthogonality. Applying the linear functional $(s, t) \mapsto t - cs$ blockwise to the first k block columns reads off the pivot coefficients; after they vanish, the remaining relation among the last r rows is killed by the invertibility of D . Thus all $k + r$ rows are independent. Their span is a totally isotropic subspace of half the ambient dimension, hence is self-dual.

For the recursive assertion, put $P' = P_{I,I}$, $H' = H_{I,*}$, $Q' = Q_{I,*}$, and $A' = A_{*,I}$. The entries of the retained matrix give (5) literally. Restricting (3) to columns in I , and (4) to rows and columns in I , gives

$$A' + D(Q')^\top = 0, \quad I_I + c(P' + (P')^\top) + c(H'(Q')^\top + Q'(H')^\top) + Q'(Q')^\top = 0.$$

The products involving H' and Q' still sum over all r terminal columns. Since D is unchanged, the converse already proved applies. For $I = \emptyset$, the remaining rows are $(D_{st}(1, c))_{t=1}^r$. As $\text{rowspan } D = K^r$, their span is exactly $U_c^{(r)}$. $\square$

Thus, writing $G_{k,r}$ for the displayed matrix, each first pivot can be removed in the same way:

$$G_{k,r} = \left[\begin{array}{c|c} (P_{11}, cP_{11} + 1) & \rho_1 \\ \hline \gamma_1(1, c) & G_{k-1,r} \end{array} \right], \quad G_{k,r} \longrightarrow G_{k-1,r} \longrightarrow \cdots \longrightarrow G_{0,r} = (D_{st}(1, c)).$$

Here $\rho_1 = (P_{12}(1, c), \dots, P_{1k}(1, c), L_1)$ and $\gamma_1 = (P_{21}, \dots, P_{k1}, A_{11}, \dots, A_{r1})^\top$. Each arrow deletes a pivot row together with its two-coordinate block column; it is not puncturing the whole code. Each retained row space is self-dual. This is the repeated boxed structure, with the same nonsingular terminal block at every step.

The integer r records $C \cap U_c$ for the fixed coordinate pairing. The entries of D , however, depend on the chosen basis of $C \cap U_c$: multiplication on the left by $\text{diag}(I_k, D^{-1})$ replaces (A, D) by $(D^{-1}A, I_r)$ without changing the code. Universality therefore requires variable r and the unrestricted pivot and terminal coefficients, not a nonidentity choice of D .

Lemma 3.1 (Building-up with fixed terminal matrix) *Let P, H, Q, A, D satisfy the hypotheses of the converse in Theorem 3.6. Choose any row vectors $h, q \in K^r$ and $u \in K^k$, and put*

$$\begin{aligned} p &= \frac{c}{2}(1 + qq^\top) - hq^\top, \\ v &= cQq^\top - Hq^\top - Qh^\top - u^\top. \end{aligned}$$

Then the coefficients

$$P^+ = \begin{pmatrix} p & u \\ v & P \end{pmatrix}, \quad H^+ = \begin{pmatrix} h \\ H \end{pmatrix}, \quad Q^+ = \begin{pmatrix} q \\ Q \end{pmatrix}, \quad A^+ = \begin{pmatrix} -Dq^\top & A \end{pmatrix}$$

and the same D define a matrix $G_{k+1,r}$ whose row space is self-dual. Deleting its first row and first two-coordinate block column gives $G_{k,r}$ exactly. This construction may be iterated arbitrarily many times, keeping r and D fixed.

Proof. Since K has odd characteristic, 2 is invertible. First,

$$A^+ + D(Q^+)^\top = \begin{pmatrix} 0 & A + DQ^\top \end{pmatrix} = 0.$$

For the other Gram relation, use its equivalent symmetric-part identity. The old lower-right block is unchanged, while the new diagonal and off-diagonal blocks satisfy

$$2p = c(1 + qq^\top) - 2hq^\top, \quad v + u^\top = cQq^\top - Hq^\top - Qh^\top.$$

Consequently,

$$P^+ + (P^+)^\top = c(I_{k+1} + Q^+(Q^+)^\top) - (H^+(Q^+)^\top + Q^+(H^+)^\top).$$

Theorem 3.6 now gives independence and self-duality. Restricting to the old rows and paired columns recovers P, H, Q, A, D entry by entry. The resulting coefficients again satisfy all the hypotheses, so the construction can be repeated. Finally, the defect map on the $(k+1+r)$ -dimensional code has rank $k+1$, so its kernel has dimension r . $\square$

The next two lemmas identify precisely how this repeated construction corresponds to a Kim–Lee step with the same parent matrix. All changes below are invertible row operations; the coordinate order and Euclidean form remain fixed.

Lemma 3.2 (Repeated steps and Kim–Lee building-up) *Use the notation and hypotheses of Lemma 3.1, and write $G = G_{k,r}$, with rows $g_1, \dots, g_m$, where $m = k + r$. Put*

$$\gamma = \begin{pmatrix} v \\ -Dq^\top \end{pmatrix}, \quad \rho = \left((u_j, cu_j)_{j=1}^k, (h_t, ch_t + q_t)_{t=1}^r \right).$$

Then the successor is exactly

$$G^+ = G_{k+1,r} = \begin{pmatrix} p & cp + 1 & \rho \\ \gamma_1 & c\gamma_1 & g_1 \\ \vdots & \vdots & \vdots \\ \gamma_m & c\gamma_m & g_m \end{pmatrix}. \quad (6)$$

Its row space has a Kim–Lee presentation with this parent and these first two coordinates if and only if $\gamma \neq 0$, equivalently,

$$\neg(q = 0 \text{ and } u^\top = -Qh^\top).$$

More precisely, if $\gamma_s \neq 0$, set

$$\lambda = \frac{c - p}{\gamma_s}, \quad x = c^{-1}(\rho + \lambda g_s). \quad (7)$$

Replacing the first row b_0 of G^+ by $c^{-1}(b_0 + \lambda b_s)$, and leaving all lower rows unchanged, gives the exact matrix

$$\begin{pmatrix} 1 & 0 & x \\ -x \cdot g_1 & -c(x \cdot g_1) & g_1 \\ \vdots & \vdots & \vdots \\ -x \cdot g_m & -c(x \cdot g_m) & g_m \end{pmatrix}, \quad x \cdot x = -1. \quad (8)$$

If $\gamma = 0$, instead,

$$p = c/2, \quad \rho \in \text{row}(G), \quad \text{row}(G^+) = K(1, -c) \oplus \text{row}(G)$$

in the displayed coordinates. This last code is not the row space of (8) for any x with this fixed parent and coordinate order.

Proof. The coefficient formulas in Lemma 3.1 give (6) entry by entry. Since G and G^+ generate self-dual codes, their rows are pairwise orthogonal. Using $c^2 = -1$, we obtain

$$\rho \cdot g_i = -c\gamma_i, \quad \rho \cdot \rho = -1 - 2cp. \quad (9)$$

Indeed, the first two coordinates contribute $c\gamma_i$ to $b_0 \cdot b_i$, and $1 + 2cp$ to $b_0 \cdot b_0$. If $\gamma_s \neq 0$, then

$$p + \lambda\gamma_s = c, \quad cp + 1 + \lambda c\gamma_s = c^2 + 1 = 0.$$

Thus the new first pair is $(1, 0)$. Moreover,

$$x \cdot g_i = c^{-1}(\rho \cdot g_i + \lambda g_s \cdot g_i) = -\gamma_i.$$

The new first row is isotropic, being a linear combination of orthogonal rows, so $1 + x \cdot x = 0$. This proves the literal equality (8). The row operation has determinant $c^{-1} \neq 0$, hence preserves the row space, and every parent tail is unchanged. In the sign convention of Theorem 3.3, the square root used in this Kim–Lee matrix is $-c$.

If $\gamma = 0$, (9) gives $\rho \in \text{row}(G)^\perp = \text{row}(G)$. Consequently $\rho \cdot \rho = 0$, and $1 + 2cp = 0$ gives $p = c/2$. Subtracting a combination of the lower rows removes ρ from b_0 . Its remaining first pair is $(c/2, 1/2) = (c/2)(1, -c)$, proving the direct-sum equality. Every vector in this direct sum has its first pair in $K(1, -c)$, which does not contain $(1, 0)$. The first row of (8) therefore cannot belong to this code. Finally, D is invertible, so $Dq^\top = 0$ forces $q = 0$; with $q = 0$, the equality $v = 0$ is precisely $u^\top = -Qh^\top$. $\square$

Lemma 3.3 (Kim–Lee steps in repeated form) *Fix a parent $G = G_{k,r}$ satisfying the hypotheses of Lemma 3.1. Every $x \in K^{2(k+r)}$ with $x \cdot x = -1$ gives a Kim–Lee matrix (8) that can be changed, by an invertible operation on its first row alone, into a successor from Lemma 3.1. The parent matrix and D are unchanged, and the resulting γ is nonzero.*

Proof. Write $\gamma_i = -x \cdot g_i$ and $b_i = (\gamma_i, c\gamma_i, g_i)$. For the old pivot blocks put

$$a_j = c(x_{2j} - cx_{2j-1}) \quad (1 \leq j \leq k), \quad b'_0 = c(1, 0, x) - \sum_{j=1}^k a_j b_j.$$

Its tail and first pair are

$$\rho = cx - \sum_{j=1}^k a_j g_j, \quad (p, cp + 1), \quad p = c - \sum_{j=1}^k a_j \gamma_j.$$

The old pivot blocks satisfy $(g_i)_{2j} - c(g_i)_{2j-1} = \delta_{ij}$ for $1 \leq i, j \leq k$. Hence

$$\rho_{2j} - c\rho_{2j-1} = c(x_{2j} - cx_{2j-1}) - a_j = 0 \quad (1 \leq j \leq k).$$

Set $u_j = \rho_{2j-1}$; in each terminal block write $\rho = (h_t, ch_t + q_t)$. Orthogonality of b'_0 to the terminal rows and then to the pivot rows gives, respectively,

$$\gamma_{k+s} = -\sum_t q_t D_{st}, \quad \gamma_i = c \sum_t q_t Q_{it} - \sum_t (h_t Q_{it} + q_t H_{it}) - u_i.$$

Finally, isotropy of b'_0 gives

$$0 = 1 + 2cp + 2c \sum_t h_t q_t + \sum_t q_t^2, \quad p = \frac{c}{2} \left(1 + \sum_t q_t^2 \right) - \sum_t h_t q_t.$$

These are exactly the successor coefficients of Lemma 3.1. No lower row has changed, and the first-row operation has determinant $c \neq 0$. If $\gamma = 0$, then x is orthogonal to every row of G , so self-duality gives $x \in \text{row}(G)$ and $x \cdot x = 0$, contradicting $x \cdot x = -1$. $\square$

Thus, with the parent and coordinate pair fixed, repeated extensions are exactly Kim–Lee steps together with the length-two direct-sum additions identified above. Different choices of s in (7) need not give the same vector x ; the statements concern generator presentations, not a bijection of parameter tuples. The direct-sum exception has minimum distance at most two, and so is excluded when the child has $d > 2$. A different coordinate permutation can change the parent reduction; no obstruction to such a different presentation is asserted.

Listing 8: Exact fixed-parent criterion in Lemma 3.2.

```

theorem repeated_step_kim_lee_iff {k r : Nat} (c : K)
  (P : Fin k -> Fin k -> K) (H Q : Fin k -> Fin r -> K)
  (A : Fin r -> Fin k -> K) (D : Fin r -> Fin r -> K)
  (hc : c ^ 2 = (-1 : K)) (h2 : Not ((2 : K) = 0))
  (hD : RankBoxCoreFullRank D)
  (hpm : PivotMasterRelations Q A D)
  (hpp : PivotGramRelations c P H Q)
  (h q : Fin r -> K) (u : Fin k -> K) :
  Iff
    (Exists fun x => And (dot x x = -1)
      (rowSpace (readSuccessor (extendedRows c P H Q A D h q u)) =
        rowSpace (buildRows x c (flattenRows (rankBoxedRows c P H Q A D))))))
    (Not (And (q = 0)
      (forall i, u i = -(Finset.univ.sum fun t => Q i t * h t))))

```

Here the paired coordinates are listed consecutively by `flattenRows`, and `readSuccessor` places the new pair first. The independent formal statements also include the explicit matrix equality in (7), the direct-sum classification, and the reverse first-row operation in Lemma 3.3.

Example 3.1 (Rank-two building-up over $\text{GF}(5)$) Take $c = 2$, so $c^2 = -1$, and let C be generated by

$$G_0 = \begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 1 & 4 & 4 & 3 & 2 \\ 1 & 4 & 3 & 4 & 2 & 3 & 4 & 2 \\ 1 & 2 & 2 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}.$$

All calculations in this example are in $\text{GF}(5)$. One checks that $G_0 G_0^T = 0$, while the minor formed by the first four columns has determinant 4. Hence C has dimension 4 and is self-dual. None of the entries of G_0 is zero, and the repeated block structure is not visible in this presentation.

Pair adjacent coordinates and let G_0^c be obtained by ordering the four two-coordinate blocks as $(2, 4, 1, 3)$. The invertible row transformation

$$T = \begin{pmatrix} 4 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 \\ 1 & 3 & 0 & 1 \\ 4 & 0 & 3 & 4 \end{pmatrix}, \quad \det T = 3,$$

gives the following matrix, written in two-coordinate blocks:

$$G_{2,2} = TG_0^\sigma = \left[\begin{array}{cc|cc} (4,4) & (2,4) & (2,1) & (3,3) \\ (4,3) & (3,2) & (2,3) & (0,2) \\ \hline (1,2) & (4,3) & (1,2) & (1,2) \\ (4,3) & (2,4) & (1,2) & (2,4) \end{array} \right].$$

This is (1) with

$$P = \begin{pmatrix} 4 & 2 \\ 4 & 3 \end{pmatrix}, \quad H = \begin{pmatrix} 2 & 3 \\ 2 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 2 & 2 \\ 4 & 2 \end{pmatrix},$$

$$A = \begin{pmatrix} 1 & 4 \\ 4 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \det D = 1.$$

For example, the four upper-left blocks have first coordinates 4, 2, 4, 3, which are exactly the entries of P . Its diagonal blocks are $(4, 2 \cdot 4 + 1) = (4, 4)$ and $(3, 2 \cdot 3 + 1) = (3, 2)$, whereas its off-diagonal blocks are $2(1, 2)$ and $4(1, 2)$. In particular, P is neither symmetric nor triangular and has nonzero diagonal. The displayed coefficients satisfy (3)–(4).

Applying $(s, t) \mapsto t - 2s$ to each block of $G_{2,2}$ gives

$$\partial G_{2,2} = \begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Its rank is 2, so $r = \dim(C \cap U_2) = 4 - 2 = 2$. Here the block permutation preserves U_2 , so the same value holds in the original coordinates. Consequently, whole-block permutations and row operations alone cannot turn this example into a rank-one form for this fixed pairing.

Now delete the first pivot row together with the first block column, and repeat once more. The retained matrices are

$$G_{1,2} = \left[\begin{array}{c|cc} (3,2) & (2,3) & (0,2) \\ \hline (4,3) & (1,2) & (1,2) \\ (2,4) & (1,2) & (2,4) \end{array} \right], \quad G_{0,2} = \begin{pmatrix} (1,2) & (1,2) \\ (1,2) & (2,4) \end{pmatrix}.$$

Theorem 3.6 gives a sequence of Euclidean self-dual row spaces with parameters

$$[8, 4] \longrightarrow [6, 3] \longrightarrow [4, 2].$$

The same terminal matrix D remains throughout, and $\text{rowspan } G_{0,2} = K(1, 2) \oplus K(1, 2)$. Each step retains only the remaining generator rows before deleting their first two coordinates; it is not puncturing the whole preceding code.

Starting instead from the same $[8, 4]$ matrix $G_{2,2}$, we can build up twice to length 12. In Lemma 3.1, choose

step	h	q	u
$G_{2,2} \longrightarrow G_{3,2}$	$(1, 2)$	$(2, 1)$	$(1, 2)$
$G_{3,2} \longrightarrow G_{4,2}$	$(2, 4)$	$(1, 3)$	$(1, 2, 4)$

with each new pivot prepended. Both steps give $p = 2$. The final matrix is

$$G_{4,2} = \left[\begin{array}{cc|cc|cc} (2,0) & (1,2) & (2,4) & (4,3) & (2,0) & (4,1) \\ (4,3) & (2,0) & (1,2) & (2,4) & (1,4) & (2,0) \\ \hline (1,2) & (3,1) & (4,4) & (2,4) & (2,1) & (3,3) \\ (3,1) & (1,2) & (4,3) & (3,2) & (2,3) & (0,2) \\ \hline (1,2) & (2,4) & (1,2) & (4,3) & (1,2) & (1,2) \\ (3,1) & (1,2) & (4,3) & (2,4) & (1,2) & (2,4) \end{array} \right].$$

The first two rows and block columns are the new ones. Their removal leaves exactly the earlier $G_{2,2} = TG_0^\sigma$, not a different $[8, 4]$ example. Removing only the first row and first block column gives $G_{3,2}$. Thus, with arrows denoting building-up and its inverse row-and-block reduction, we have

$$G_{2,2} ([8, 4]) \longleftrightarrow G_{3,2} ([10, 5]) \longleftrightarrow G_{4,2} ([12, 6]).$$

All three row spaces are Euclidean self-dual, and all have $r = 2$ and the same terminal matrix D .

These two extensions also give the literal Kim–Lee construction, not just matrices satisfying the Gram equations. Multiplying the first row of each successor by $3 = 2^{-1}$ changes its first block to $(1, 0)$. The corresponding tail vectors are

$$x_1 = (3, 1, 1, 2, 3, 2, 1, 0), \quad x_2 = (3, 1, 1, 2, 2, 4, 1, 0, 2, 3),$$

and $x_1 \cdot x_1 = x_2 \cdot x_2 = 4 = -1$. For each parent row g_i , the newly prepended pair is $(-y_i, 3y_i)$, where $y_i = x_j \cdot g_i$. Thus Theorem 3.3 applies with its square root of -1 chosen as $3 = -2$; the boxed matrices themselves use $c = 2$. Only a generator-row scaling is involved, so neither the Euclidean form nor the coordinate pairing is changed. Lemma 3.1, beyond these two numerical steps, ensures further extensions with the same D . Since these two steps have $p = c$, their row scalings are the case $\lambda = 0$ of Lemma 3.2; Lemma 3.3 supplies the converse for every Kim–Lee extension of either parent.

3.3 Binary rank-one refinement

The same matrix formula also applies over $\mathbb{F}_2$. For positive length, Theorem 3.8 gives a scalar-coordinate permutation with $r = 1$, $D = (1)$, zero pivot diagonal, and $H_{i1} = Q_{i1} = A_{1i} = 1$. With $c = 1$, the terminal pivot block is then $(1, 1 + 1) = (1, 0)$, so (1) is the literal binary box. This permutation may change the original pairing of coordinates; it does not assert $r = 1$ for every fixed pairing.

The normalization in (7) does not use division by two. In a normalized binary CZ step, $c = 1$, $p = 0$, and the all-ones last row has $\gamma_s = 1$. Thus $\lambda = 1$, $x = \rho + \mathbf{1}$, and the first-row operation is exactly $b_0 \mapsto b_0 + b_s$. It gives $x \cdot x = 1$ and the binary Kim matrix, with the same parent tails. The nonzero last entry of γ excludes the direct-sum exception at every

such step. This specialization uses the first-row normalization, not the odd-characteristic coefficient formula involving $c/2$.

In 2012, Chinburg and Zhang [5] proved that up to permutation equivalence, all binary self-dual codes of length at least 4 arise from Hilbert pairings on rings of S -integers of $\mathbb{Q}$.

Theorem 3.7 (*[5, Theorem 1.5]*) *Up to permutation equivalence, all binary self-dual codes of length at least 4 arise from Hilbert pairings on rings of S -integers of $K = \mathbb{Q}$. In fact, each such code arises up to permutation equivalence from infinitely many different subsets S of the places of $K = \mathbb{Q}$.*

Here “equivalence” is used exactly in the sense defined by Chinburg–Zhang: a bijection of the chosen orthonormal bases whose linear extension carries one code to the other. Thus it is coordinate-permutation equivalence, not arbitrary monomial equivalence.

In the course of their proof, Chinburg–Zhang exhibit a boxed normal form for the resulting binary self-dual code. In this form, the bottom row is the all-ones vector, the distinguished terminal block-column is of type 10 above the bottom row, the diagonal block-entries are of type 01, and every remaining off-diagonal block is either 00 or 11, paired so that the two opposite blocks sum to 11.

In the table below, $W_v = \mathbb{Q}_v^\times / (\mathbb{Q}_v^\times)^2$ is the local square-class space with its Hilbert pairing. The primes p_i are distinct and congruent to 3 (mod 4); the column headings list local bases and the rows list global S -units [5].

S -units $\setminus$ places	W_{p_1} $\{-p_1, p_1\}$	W_{p_2} $\{-p_2, p_2\}$	$\dots$ $\dots$	W_2 $\{-2, -10, -5\}$	$W_{\mathbb{R}}$ $\{-1\}$	
p_1	01	00/11		00/11	1	0
p_2	11/00	01		00/11	1	0
$\vdots$	$\vdots$		$\ddots$			$\vdots$
2	11/00	11/00		01	1	0
-1	11	11	$\dots$	11	1	1

The point of this boxed form is twofold: every binary self-dual code of length $2n$ admits such a representative, and deleting the distinguished two-coordinate block together with the top row produces a shorter self-dual code of length $2n - 2$. This is a coordinate reduction in the standard Euclidean form. The deleted coordinate plane has Gram matrix I_2 ; it is not an alternating hyperbolic plane.

The two-coordinate reduction itself does not require the entire matrix to have already been put in boxed form. It retains the codewords whose first two coordinates are equal and then deletes those coordinates. This differs from ordinary coordinate shortening, which retains only codewords that are zero at the deleted positions. The following lemma gives the reduction used in the induction.

Lemma 3.4 (Binary two-coordinate reduction) *Let $C = C^\perp \subseteq \mathbb{F}_2^{2+s}$, and suppose that $x \in C$ satisfies $x_1 = 0$ and $x_2 = 1$. Put*

$$C_- := \{v \in C : v_1 + v_2 = 0\}, \quad \pi(v_1, v_2, v') = v', \quad D := \pi(C_-) \subseteq \mathbb{F}_2^s.$$

Then $\pi|_{C_-}$ is injective and $D = D^\perp$. In particular, s is even and $\dim D = s/2$.

Proof. If $v \in C_-$ and $\pi(v) = 0$, then $v = (a, a, 0)$. Since $v, x \in C$ and $C \subseteq C^\perp$,

$$0 = x \cdot v = 0 \cdot a + 1 \cdot a = a.$$

Thus $v = 0$, proving injectivity. If $u = (a, a, g)$ and $v = (b, b, h)$ belong to C_- , then

$$0 = u \cdot v = 2ab + g \cdot h = g \cdot h,$$

so $D \subseteq D^\perp$. Finally, the functional $\delta : C \rightarrow \mathbb{F}_2$, $\delta(v) = v_1 + v_2$, is surjective because $\delta(x) = 1$. Hence

$$\dim C_- = \dim C - 1 = \frac{2+s}{2} - 1 = \frac{s}{2}.$$

Injectivity of $\pi|_{C_-}$ gives $\dim D = s/2$; together with $D \subseteq D^\perp$, this yields $D = D^\perp$. $\square$

In the following listing, the restriction $\pi|_{C_-}$ is denoted by `binaryTwoCoordinateMap`. Its image D is denoted by `binaryReducedCode`. The conclusions state injectivity, self-duality, the dimension formula, and the evenness of the length.

Listing 9: Lean formulation of binary two-coordinate reduction.

```
theorem paper_binary_twoCoordinateReduction_exact {n : ℕ}
  {C : Submodule (ZMod 2) (Fin (2 + n) → ZMod 2)}
  {x : Fin (2 + n) → ZMod 2}
  (hC : paperSelfDualCode (K := ZMod 2) C)
  (hx0 : x 0 = 0) (hx1 : x 1 = 1) :
  Function.Injective (binaryTwoCoordinateMap C) ∧
  paperSelfDualCode (K := ZMod 2) (binaryReducedCode C) ∧
  2 * Module.finiteRank (ZMod 2) (binaryReducedCode C) = n ∧ Even n
```

Theorem 3.8 (Universal binary rank-one boxed normal form) *For every $N \geq 1$ and every Euclidean self-dual code $C \subseteq \mathbb{F}_2^{2N}$, there are a permutation σ of the $2N$ scalar coordinates and a matrix $b = (b_{ij}) \in M_{N-1}(\mathbb{F}_2)$ such that*

$$b_{ii} = 0, \quad b_{ij} + b_{ji} = 1 \quad (i \neq j). \quad (10)$$

and $\sigma(C) = \text{rowspan } G(b)$, where, writing $a(11) = (a, a)$,

$$G(b) = \left[\begin{array}{cccc|c} 01 & b_{12}(11) & b_{13}(11) & \cdots & b_{1,N-1}(11) & 10 \\ b_{21}(11) & 01 & b_{23}(11) & \cdots & b_{2,N-1}(11) & 10 \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ b_{N-1,1}(11) & b_{N-1,2}(11) & \cdots & & 01 & 10 \\ \hline 11 & 11 & \cdots & & 11 & 11 \end{array} \right]. \quad (11)$$

For $N = 1$, this means the single row 11.

Proof. We induct on N . If $N = 1$, self-duality forces C to be the unique isotropic line $\langle(1, 1)\rangle \subseteq \mathbb{F}_2^2$. Assume $N \geq 2$. For every $v \in C$,

$$v \cdot \mathbf{1} = v \cdot v = 0;$$

hence $\mathbf{1} \in C^\perp = C$. Since $\dim C = N > 1$, some word of C is nonconstant. After a scalar-coordinate permutation it has the form $x = (0, 1, x')$. The binary two-coordinate reduction lemma gives the self-dual predecessor

$$D = \pi\{v \in C : v_1 + v_2 = 0\} \subseteq \mathbb{F}_2^{2N-2}.$$

By induction, after a permutation of the tail coordinates, D is generated by a box $G(b)$ with pivot rows $g_1, \dots, g_{N-2}$ and all-ones row $M = \mathbf{1}$, satisfying (10).

Put $p = x + \mathbf{1} = (1, 0, Z)$. Since $p \in C = C^\perp$,

$$0 = p \cdot p = 1 + Z \cdot Z, \quad Z \cdot Z = 1. \quad (12)$$

Exact reconstruction from the two-coordinate reduction gives

$$C = \text{span} \left\{ A = (1, 0, Z), \quad \hat{g}_i = (Z \cdot g_i, Z \cdot g_i, g_i), \quad \widehat{M} = (Z \cdot M, Z \cdot M, M) \right\}. \quad (13)$$

Thus this is literally the Kim extension of the inductively normalized predecessor.

For an old two-coordinate pivot block j , let $\delta_j(Z) = Z_{j,1} + Z_{j,2}$, and set

$$U = A + \sum_{j=1}^{N-2} \delta_j(Z) \hat{g}_j, \quad T = U + U_{\text{term},2} \widehat{M}. \quad (14)$$

In the old box, the linear functional $(u, v) \mapsto u + v$ is one on the diagonal block of g_j , one on its terminal block, and zero on every other pivot block and on M . Consequently every old pivot block of T has coordinate sum zero and is therefore either 00 or 11. Equation (12) gives

$$T_{\text{term}} = 10. \quad (15)$$

If $s = T_{1,2}$, then the new diagonal block of T is $(1 + s, s)$. It is already 01 when $s = 1$, and swapping the two new scalar coordinates makes it 01 when $s = 0$. This swap fixes the equal new head blocks of every $\hat{g}_i$ and $\widehat{M}$. The latter is the all-ones row, while all old diagonal and terminal blocks remain 01 and 10, respectively. Hence these corrected rows have exactly the form (11) for a matrix b' .

Row operations and the optional pair swap preserve the code and its Euclidean inner product. For two distinct pivot rows of the resulting box, direct block multiplication gives

$$0 = g_i(b') \cdot g_j(b') = b'_{ij} + b'_{ji} + 1,$$

so $b'_{ij} + b'_{ji} = 1$; the diagonal blocks give $b'_{ii} = 0$. Composing the initial pivot permutation, the inductive tail permutation, and the optional new-pair swap yields the required single scalar-coordinate permutation σ . $\square$

Listing 10: Lean statement for the universal binary rank-one form.

```

theorem binarySelfDualCode_has_rankOneNormalForm
  {k : Nat}
  {C : Submodule (ZMod 2) (Fin (2 * (k + 1))) → ZMod 2)}
  (hC : paperSelfDualCode C) :
  HasBinaryCzRankOneNormalForm C

```

The similarity between Theorem 3.1 and this boxed matrix is not accidental. Kim’s theorem is a bottom-up extension statement, while the Chinburg–Zhang boxed form is a top-down reduction statement. The next theorem shows that the extension and reduction are inverse constructions.

For the arithmetic comparison, let K be a number field, let S be a finite set containing its archimedean and dyadic places, and put $U = \text{Spec } \mathcal{O}_{K,S}$. Set

$$W = \bigoplus_{v \in S} H_{\text{et}}^1(K_v, \mu_2), \quad \Phi : H_{\text{et}}^1(U, \mu_2) \longrightarrow W, \quad L = \text{im } \Phi.$$

At real places the local groups use Tate cohomology. The pairing B_{coh} is the sum of the local Hilbert pairings, equivalently the cup product followed by the compact-support boundary map [5, Theorem 1.1]; it is nondegenerate and $L = L^{\perp_{B_{\text{coh}}}}$. If -1 is nonsquare in every non-complex K_v for $v \in S$, local orthonormal bases give an ordered basis E of W [5, Theorem 1.2 and Corollary 1.4]. Writing $\dim W = 2n$, its coordinate map is a form isometry

$$\iota_E : (W, B_{\text{coh}}) \xrightarrow{\sim} (\mathbb{F}_2^{2n}, \langle \ , \ \rangle_E).$$

The following comparison uses only these linear-algebraic data.

Theorem 3.9 (Cohomological transport and binary boxed reduction) *Let $n \geq 2$, and let $L = \text{im } \Phi \subset W$ be the Chinburg–Zhang arithmetic image and suppose*

$$L = L^{\perp_{B_{\text{coh}}}}.$$

Fix the form isometry ι_E above and put $C = \iota_E(L)$. Apply Theorem 3.8 and add the all-ones row to the first pivot row. The resulting permutation-equivalent representative of C has generator matrix

$$G' = \begin{pmatrix} 1 & 0 & x \\ y_1 & y_1 & g_1 \\ \vdots & \vdots & \vdots \\ y_{n-1} & y_{n-1} & g_{n-1} \end{pmatrix}, \quad x, g_i \in \mathbb{F}_2^{2n-2}.$$

Let G be the matrix with rows $g_1, \dots, g_{n-1}$ and let $C' = \text{rowspan}(G)$. Then:

1. L is self-orthogonal for B_{coh} and C is Euclidean self-dual.

2. In the displayed Euclidean coordinates,

$$\mathbb{F}_2^{2n} = E_2 \perp \mathbb{F}_2^{2n-2}, \quad \text{Gram}(E_2) = I_2.$$

The plane E_2 is non-alternating and is not isometric to the alternating hyperbolic plane. Nevertheless, $\langle(1, 1)\rangle \subset E_2$ is an isotropic line and $\langle(1, 0), (1, 1)\rangle_E = 1$.

3. The rows g_i are linearly independent and pairwise orthogonal; hence C' is a Euclidean self-dual code of length $2n - 2$.

4. For every i ,

$$y_i = \langle x, g_i \rangle_E.$$

Consequently G' is exactly the Kim building-up matrix obtained from G and x .

5. Deleting the top row and the first two coordinates of G' gives G , and rebuilding from G gives G' exactly. Therefore, after restoring the coordinate permutation used to obtain the boxed representative, the Chinburg–Zhang boxed reduction and the Kim extension are inverse up to permutation equivalence.

Proof. The assumed equality $L = L^{\perp_{B_{\text{coh}}}}$ implies self-orthogonality. Since ι_E preserves the two forms, it transports orthogonal complements, so $C = \iota_E(L)$ is Euclidean self-dual. Theorem 3.8 gives a boxed representative whose first pivot row starts with 01 and whose all-ones row starts with 11. Adding the latter to the former changes that head to 10. Every other row has equal first coordinates. This invertible row operation gives precisely the displayed matrix G' , without any further change of coordinates. Composing ι_E with the selected coordinate permutation gives a form isometry whose image of L is exactly the displayed row space.

The first two standard coordinate vectors have Gram matrix I_2 . Thus E_2 is non-alternating because $(1, 0)$ has self-product 1. The alternating hyperbolic plane has Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and alternation is invariant under form isometry, so these planes are not isometric. On the other hand,

$$(1, 1) \cdot (1, 1) = 0, \quad (1, 0) \cdot (1, 1) = 1,$$

which gives the stated isotropic line in E_2 without turning E_2 into a hyperbolic plane.

Self-duality of C gives orthogonality of the rows of G' . In particular, $0 = (1, 0, x) \cdot (1, 0, x) = 1 + x \cdot x$, so $x \cdot x = 1$. Moreover,

$$0 = (1, 0, x) \cdot (y_i, y_i, g_i) = y_i + x \cdot g_i,$$

and characteristic two yields $y_i = x \cdot g_i$. Therefore G' is precisely the generator family in Kim's building-up construction. Since C has dimension n and G' has n rows, those rows are linearly independent. The linear map

$$g \longmapsto (x \cdot g, x \cdot g, g)$$

has tail projection as a left inverse, so the rows of G are linearly independent. For all i, j ,

$$(y_i, y_i, g_i) \cdot (y_j, y_j, g_j) = 2y_i y_j + g_i \cdot g_j = g_i \cdot g_j.$$

Thus the g_i are pairwise orthogonal. Their span has dimension $n - 1$, half the ambient length $2n - 2$, and is therefore self-dual. Finally, tail projection gives G after deleting the top row and first coordinate pair, while the coefficient identity rebuilds G' exactly. The only remaining change from the original code is the coordinate permutation used to reach the boxed representative. $\square$

In the following listing, `e` includes the selected permutation of the orthonormal coordinates. Thus `hcode` is equality with the actual image of L . The internal predicate `IsSelfOrthogonal` denotes equality with the orthogonal complement, as required by the hypothesis on L .

Listing 11: Lean for Theorem 3.9.

```

theorem dot_r0_riBin_eq_zero_iff [CharP K 2] :
  dot (r0 x) (riBin y g) = 0 <-> y = dot x g

theorem boxedFamily_tail_paperSelfDualCode [CharP K 2]
  (hself : paperSelfDualCode (rowSpace (boxedFamily x Y G))) :
  paperSelfDualCode (rowSpace G)

theorem paper_binary_cz_kim_corrected [CharP K 2]
  (he : IsFormIsometry Bcoh dotBilin e)
  (hL : IsSelfOrthogonal Bcoh L)
  (hcode : L.map e.toLinearMap =
    rowSpace (boxedFamily x Y G)) :
  IsSelfOrthogonal dotBilin (L.map e.toLinearMap) ∧
  paperSelfDualCode (rowSpace G) ∧
  ∀( i, Y i = dot x (G i)) ∧
  boxedFamily x Y G = buildRowsBin x G ∧
  deleteBinaryHeadPair (boxedFamily x Y G) = G ∧
  CodeEquiv
  (buildRowsBin x (deleteBinaryHeadPair (boxedFamily x Y G)))
  (boxedFamily x Y G)

```

Remark 3.3 In Theorem 3.9, the first coordinate plane $E_2 = \mathbb{F}_2^2$ has Gram matrix I_2 . Its diagonal line $\mathbb{F}_2(1, 1)$ is isotropic, but the form on E_2 is non-alternating; in particular, E_2 is not an alternating hyperbolic plane.

The arithmetic code is $C = \iota_E(\text{im } \Phi)$, with Φ and ι_E defined above. The comparison with Kim's matrix takes place in these Euclidean coordinates.

The coefficient group $H_{\text{et}}^1(U, \mu_2)$ is annihilated by 2. If q is odd, every additive homomorphism $f : H_{\text{et}}^1(U, \mu_2) \rightarrow \mathbb{F}_q^{2n}$ is therefore zero:

$$2f(a) = f(2a) = 0, \quad 2 \in \mathbb{F}_q^\times.$$

Thus a nonzero odd-characteristic code cannot be obtained from this restriction image by an additive coordinate identification. In Theorem 3.5, the correction terms instead lie on $\mathbb{F}_q(1, c)$, whose isotropy follows from $c^2 = -1$.

3.4 Normalized rank-one forms over odd fields

We return to $K = \mathbb{F}_q$, $q \equiv 1 \pmod{4}$, and fix $c \in K^\times$ with $c^2 = -1$.

To recover the following rank-one form, set $k = n - 1$, $r = 1$, $D = (1)$ and $P_{ii} = 0$. The latter is an additional diagonal normalization, not a consequence of $r = 1$. The exact coefficient correspondence is

$$P_{ij} = b_{ij} \ (i \neq j), \quad H_{i1} = u_i, \quad Q_{i1} = v_i - cu_i, \quad A_{1i} = a_i. \quad (16)$$

Corollary 3.10 (Normalized rank-one split boxed form over $\mathbb{F}_q$) *Fix $c \in K^\times$ with $c^2 = -1$. For $1 \leq i \leq n - 1$, let $\ell_i = (u_i, v_i) \in K^2$ and $a_i \in K$, and for distinct $1 \leq i, j \leq n - 1$, let $b_{ij} \in K$. Let G be the $n \times 2n$ generator matrix whose $n \times n$ block form $\tilde{G} = (B_{ij})$ of 1×2 blocks is determined by*

$$B_{ii} = (0, 1) \quad (1 \leq i \leq n - 1), \quad B_{nn} = (1, c),$$

$$B_{in} = \ell_i \quad (1 \leq i \leq n - 1), \quad B_{ni} = a_i(1, c) \quad (1 \leq i \leq n - 1),$$

and

$$B_{ij} = b_{ij}(1, c) \quad (1 \leq i \neq j \leq n - 1).$$

Thus $\tilde{G}$ has the split boxed shape

$$\tilde{G} = \left[\begin{array}{c|ccccc|c} 01 & b_{12}(1, c) & b_{13}(1, c) & \cdots & b_{1,n-1}(1, c) & \ell_1 \\ \hline b_{21}(1, c) & 01 & b_{23}(1, c) & \cdots & b_{2,n-1}(1, c) & \ell_2 \\ b_{31}(1, c) & b_{32}(1, c) & 01 & \cdots & b_{3,n-1}(1, c) & \ell_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ b_{n-1,1}(1, c) & b_{n-1,2}(1, c) & b_{n-1,3}(1, c) & \cdots & 01 & \ell_{n-1} \\ \hline a_1(1, c) & a_2(1, c) & a_3(1, c) & \cdots & a_{n-1}(1, c) & (1, c) \end{array} \right],$$

where each entry is a two-coordinate block. Reading these blocks as scalar pairs gives the ordinary $n \times 2n$ matrix G . Assume that

$$u_i^2 + v_i^2 = -1 \quad (1 \leq i \leq n - 1), \quad (17)$$

$$u_i + cv_i = -ca_i \quad (1 \leq i \leq n - 1), \quad (18)$$

and

$$c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j = 0 \quad (1 \leq i < j \leq n-1), \quad (19)$$

where $\ell_i \cdot \ell_j = u_i u_j + v_i v_j$ is the Euclidean dot product on the distinguished 2-plane. Then the rows of G are pairwise orthogonal and linearly independent. Therefore G generates a self-dual code of length $2n$.

Proof. Use the parameters in (16), with $P_{ii} = 0$. Since

$$(u_i, cu_i + (v_i - cu_i)) = (u_i, v_i) = \ell_i,$$

the matrix in the corollary is literally $G(c; P, H, Q, A, (1))$, with the final row and block column placed last. Write $Q_i = v_i - cu_i$. The equation in (3) becomes $a_i + Q_i = 0$; because $c \neq 0$ and $c^2 = -1$, this is equivalent to

$$c(a_i + Q_i) = ca_i + u_i + cv_i = 0,$$

which is (18). The diagonal entry of (4) is

$$1 + 2cu_i Q_i + Q_i^2 = 1 + u_i^2 + v_i^2,$$

so its vanishing is exactly (17). For $i \neq j$, the corresponding entry is

$$c(b_{ij} + b_{ji}) + c(u_i Q_j + Q_i u_j) + Q_i Q_j = c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j.$$

Symmetry reduces these equations to $i < j$, giving precisely (19). Thus the three hypotheses are equivalent to the two universal Gram relations, and $\det(1) = 1$. Theorem 3.6 gives all three conclusions.

For clarity, the two non-final mixed contributions for rows $i < j$ are

$$(0, 1) \cdot b_{ji}(1, c) + b_{ij}(1, c) \cdot (0, 1) = cb_{ji} + cb_{ij};$$

all other non-final off-diagonal contributions vanish since $(1, c) \cdot (1, c) = 0$. Linear independence also follows directly without any Gram hypothesis. Writing g_i for the rows of G , suppose that $\sum_i \lambda_i g_i = 0$ and apply the linear functional $\beta(s, t) = t - cs$ to each non-final block column j . Only the diagonal block $(0, 1)$ survives, so $\lambda_j = 0$. The first coordinate of the final block then gives $\lambda_n = 0$. $\square$

In Lean, $m = n-1$, the row and block index type is `Option (Fin m)`, and the split block-row space is canonically $K^{2(m+1)} = K^{2n}$. The following formulation states pairwise orthogonality, linear independence, and self-duality under the three displayed coefficient conditions.

Listing 12: Lean for Corollary 3.10.

```

theorem paper_split_boxed_form_exact
  (c : K) (ell : Fin m -> SplitBlock K)
  (a : Fin m -> K) (b : Fin m -> Fin m -> K)
  (hc : c ^ 2 = (-1 : K))
  (hnorm : ∀ i, dot (ell i) (ell i) = (-1 : K))
  (hlast : ∀ i, ell i 0 + c * ell i 1 = -c * a i)
  (hoff : ∀ i j, i < j ->
    c * (b i j + b j i) + dot (ell i) (ell j) = 0) :
  SplitBoxedPairwiseOrthogonal (splitBoxedRows c ell a b) ∧
  LinearIndependent K (splitBoxedRows c ell a b) ∧
  splitBoxedRowSpace (splitBoxedRows c ell a b) =
    (splitBlockRowBilin (K := K) (m := m)).orthogonal
    (splitBoxedRowSpace (splitBoxedRows c ell a b))

```

Conversely, self-duality forces the coefficient relations of Corollary 3.10 whenever a generator matrix satisfies the following block conditions.

Theorem 3.11 (Conditional boxed normalization from a distinguished isotropic-line row)
Let $n \geq 1$, let $C \subset K^{2n}$ be a self-dual code, and fix $c \in K^\times$ with $c^2 = -1$. Assume that, after coordinate permutations preserving the 2-block decomposition (hence Euclidean isometries) and elementary row operations, C admits a generator matrix of the following form. Work in these coordinates, and continue to denote the code by C . Its last row g_n is

$$(a_1(1, c), a_2(1, c), \dots, a_{n-1}(1, c), (1, c)).$$

The remaining rows $g_1, \dots, g_{n-1}$ satisfy the following block conditions:

1. *the i -th block of g_i is 01;*
2. *the last block of g_i is $\ell_i = (u_i, v_i)$;*
3. *every other non-final block of g_i lies on the same c -isotropic line, so for each $j \neq i, n$ one has*

$$B_{ij} = b_{ij}(1, c)$$

for some scalar $b_{ij} \in K$.

Then the coefficients satisfy

$$u_i^2 + v_i^2 = -1, \quad u_i + cv_i = -ca_i, \quad c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j = 0,$$

where the first two relations hold for $1 \leq i \leq n-1$ and the third for $1 \leq i < j \leq n-1$. Thus the generator matrix is the matrix of Corollary 3.10, with these coefficient relations.

Proof. Write the adapted generator matrix in block form as

$$\tilde{G} = (B_{ij})_{1 \leq i, j \leq n},$$

with $B_{ii} = 01$ for $1 \leq i \leq n-1$, $B_{in} = \ell_i$, $B_{ni} = a_i(1, c)$, and $B_{ij} = b_{ij}(1, c)$ for $i \neq j < n$. Since C is self-dual, its generator rows are pairwise orthogonal.

Applying self-orthogonality to a non-final row g_i gives

$$0 = \langle g_i, g_i \rangle = 1 + (u_i^2 + v_i^2),$$

because the diagonal block contributes 1, the final block contributes $u_i^2 + v_i^2$, and every other block is isotropic. Hence

$$u_i^2 + v_i^2 = -1.$$

Next, orthogonality of g_i with the distinguished last row gives

$$0 = \langle g_i, g_n \rangle = (0, 1) \cdot a_i(1, c) + \ell_i \cdot (1, c) = ca_i + (u_i + cv_i),$$

so

$$u_i + cv_i = -ca_i.$$

Finally, for $1 \leq i < j \leq n-1$, orthogonality of g_i and g_j gives

$$\begin{aligned} 0 = \langle g_i, g_j \rangle &= (0, 1) \cdot b_{ji}(1, c) + b_{ij}(1, c) \cdot (0, 1) + \ell_i \cdot \ell_j \\ &= c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j, \end{aligned}$$

where all other block products vanish because $(1, c) \cdot (1, c) = 0$. These are exactly the boxed relations in Corollary 3.10. $\square$

In the following formulation, $m = n-1$, and **none** denotes the last row or block. The bilinear form is the Euclidean dot product in block notation. The hypotheses specify the generator row space and every block of the matrix; the conclusions give that matrix and its coefficient relations.

Listing 13: Lean formulation of Theorem 3.11.

```

theorem paper_conditional_boxed_normalization_exact
  {m : ℕ} (c : K) (hc : c ^ 2 = (-1 : K))
  (ell : Fin m → SplitBlock K) (a : Fin m → K)
  (b : Fin m → Fin m → K)
  (G : Option (Fin m) → SplitBlockRow K m)
  (C : Submodule K (SplitBlockRow K m))
  (hC : C = (splitBlockRowBilin (K := K) (m := m)).orthogonal C)
  (hgen : splitBoxedRowSpace G = C)
  (hfinal : G none none = isotropicLineBlock c 1)
  (hfinalLeft : ∀ j, G none (some j) = isotropicLineBlock c (a j))
  (hlast : ∀ i, G (some i) none = ell i)
  (hdiag : ∀ i, G (some i) (some i) = splitDiagonalBlock)
  (hother : ∀ i j, i ≠ j →
    G (some i) (some j) = isotropicLineBlock c (b i j)) :
  G = splitBoxedRows c ell a b ∧
  ∀( i, dot (ell i) (ell i) = (-1 : K)) ∧
  ∀( i, ell i 0 + c * ell i 1 = -c * a i) ∧
  ∀( i j, i < j → c * (b i j + b j i) + dot (ell i) (ell j) = 0) ∧
  C = splitBoxedRowSpace (splitBoxedRows c ell a b)

```

Remark 3.4 Let $K = \mathbb{F}_q$, where $q \equiv 1 \pmod{4}$, choose $c \in K$ with $c^2 = -1$, and fix a pairing of the $2n$ coordinates. Put $U_c = \bigoplus_{j=1}^n K(1, c)$. For every $C = C^\perp \subseteq K^{2n}$, Theorem 3.6 gives a generator $G(c; P, H, Q, A, D)$, after a permutation of the two-coordinate blocks, with

$$r = \dim(C \cap U_c), \quad k = n - r, \quad \det D \neq 0.$$

When $r = 1$, scaling the terminal row normalizes D to (1). The further conditions $P_{ii} = 0$ are additional assumptions, not consequences of $r = 1$. Under these conditions, the substitution (16) reduces the Gram relations to (17)–(19), giving Corollary 3.10. For a generator satisfying all the block conditions of Theorem 3.11, self-duality implies precisely these three coefficient relations.

4 Applications

The applications below separate reproducible catalogue entries, repeated parent–child lineages, and the universal realization used in the main argument.

4.1 Verified application catalogue

For earlier results on self-dual codes over these fields, see [3, 10, 14]. For a displayed generator G , write

$$A_j = \#\{a \in \mathbb{F}_p^k : \text{wt}(aG) = j\}.$$

Table 1 compares our exact pair (d, A_d) with a published benchmark at the same length. The benchmark A_d is the coefficient of the cited representative code, not an extremal quantity. This distinction matters at length 12 over $\text{GF}(13)$, where inequivalent optimal codes have different weight enumerators. For MDS rows, by contrast, $A_d = (p-1)\binom{n}{d}$ is determined by the parameters.

Code	p	Our $[n, k, d]; A_d$	Published $(d_{\text{best}}; A_d)$	Reference
$C_{16}^{(2)}$	2	[16, 8, 4]; 12	(4; 12)	Huffman–Pless
$C_{18}^{(2)}$	2	[18, 9, 4]; 9	(4; 9)	Huffman–Pless
$C_{20}^{(2)}$	2	[20, 10, 4]; 5	(4; 5)	Huffman–Pless
$C_{22}^{(2)}$	2	[22, 11, 6]; 77	(6; 77)	Huffman–Pless
$\mathcal{G}_{24}$	2	[24, 12, 8]; 759	(8; 759)	Huffman–Pless
$C_{24}^{(5)}$	5	[24, 12, 9]; 1,056	(9; 1,056)	Georgiou–Koukouvinos
$C_6^{(5)}$	5	[6, 3, 4]; 60	(4; 60)	Leon–Pless–Sloane database
$C_8^{(5)}$	5	[8, 4, 4]; 48	(4; 48)	Leon–Pless–Sloane database
$C_8^{(13)}$	13	[8, 4, 5]; 672	(5; 672)	Betsumiya et al.
$C_{10}^{(13)}$	13	[10, 5, 6]; 2,520	(6; 2,520)	Betsumiya et al.
$C_{12,A}^{(13)}$	13	[12, 6, 6]; 576	(6; {528, 576, 696, 792})	Betsumiya et al.
$C_{12,P}^{(13)}$	13	[12, 6, 6]; 960	(6; {528, 576, 696, 792})	Betsumiya et al.
$C_{14}^{(13)}$	13	[14, 7, 8]; 36,036	(8; 36,036)	Betsumiya et al.
$C_{18}^{(13)}$	13	[18, 9, 7]; 72	(8; 2,484)	Betsumiya et al.
$C_{20}^{(13)}$	13	[20, 10, 8]; 120	(10; 111,024)	Betsumiya et al.
$C_{22}^{(13)}$	13	[22, 11, 10]; 23,628	(10; 23,628)	Betsumiya et al.

Table 1: Certified self-dual codes and published distance benchmarks.

Each code symbol in Table 1 links to its exact generator and verification record. The numerical verifiers check the relevant boxed or repeated relation, rank, Gram matrix, and recorded weight data; every complete distribution is also checked against the MacWilliams identity. The three largest repeated generators are written out in full below. The finite enumerations are external exact computations rather than Lean proofs.

4.2 Largest repeated boxed presentations

The same two-coordinate border displays the largest certified code over each of the three fields. If G has rows $g_1, \dots, g_m$, let

$$\mathcal{B}_c(G, x) = \begin{pmatrix} 1 & 0 & x \\ -y_1 & -cy_1 & g_1 \\ \vdots & \vdots & \vdots \\ -y_m & -cy_m & g_m \end{pmatrix}, \quad y_i = x \cdot g_i, \quad x \cdot x = -1. \quad (20)$$

For $q = 2, 5, 13$, take $c = 1, 2, 5$, respectively. In characteristic two, $-1 = 1$ and the lower heads become (y_i, y_i) , so $\mathcal{B}_1$ is literally the binary repeated box. With the coordinate orders recorded in the verification records, the three largest presentations are

$$\begin{aligned}\tilde{G}_{24}^{(2)} &= \mathcal{B}_1(G_{22}^{(2)}, x_{22}^{(2)}), \\ \tilde{G}_{24}^{(5)} &= \mathcal{B}_2(G_{22}^{(5)}, x_{22}^{(5)}), \\ \tilde{G}_{22}^{(13)} &= \mathcal{B}_5(G_{20}^{(13)}, x_{20}^{(13)}).\end{aligned}\tag{21}$$

Each equality is an exact generator identity after the certified row normalization and coordinate permutation, not only an abstract equivalence of codes.

The extended binary Golay code gives a compact instance in which two exact levels can be followed in either direction from length 20. Let $\mathcal{G}_{24}$ be the extended Golay code with its standard generator G_{24} , and put $C_{24} = \mathcal{G}_{24}$. Repeatedly apply the two-coordinate reduction of Theorem 3.9 to the first coordinate pair of the current tail. Deterministic row reduction gives self-dual generators G_m , for $m = 16, 18, 20, 22$, such that

$$\mathcal{B}_1(G_m, x_m) = \begin{pmatrix} 1 & 0 & x_m \\ y_m & y_m & G_m \end{pmatrix} \text{ generates } C_{m+2}, \quad m \in \{16, 18, 20, 22\}, \tag{22}$$

where $C_m = \text{rowspan}(G_m)$ and

$$\begin{aligned}x_{16} &= 0000001100110111, & x_{18} &= 000000110011011001, \\ x_{20} &= 00000000110100101011, & x_{22} &= 0000000000101011100011, \\ (y_m)_i &= x_m \cdot (G_m)_i.\end{aligned}$$

Here $x_{22}^{(2)} = x_{22}$ in (21). The complete normalized generator is

$$\tilde{G}_{24}^{(2)} = \left(\begin{array}{cc|cc|cc|cc|cc|cc|cc|cc|cc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \end{array} \right).$$

The vertical rules delimit the twelve two-coordinate blocks; they are not additional entries of the matrix. Thus length 20 is the center of the requested two-level picture:

$$C_{16} \longleftarrow C_{18} \longleftarrow C_{20} \longrightarrow C_{22} \longrightarrow C_{24} = \mathcal{G}_{24}.$$

Following the arrows to the left means two-coordinate reduction; the right arrows are Kim building-up with x_{20} and x_{22} . Every left edge is also inverted exactly by the corresponding Kim step with x_{16} or x_{18} , so all four adjacent pairs satisfy the same reconstruction identity. The parameters and the number A_d of minimum-weight words at each level are recorded in Table 2. In particular, the distance rises at both upward steps from the length-20 center. All five distances attain the classical upper bounds for binary self-dual codes [8].

Over $\text{GF}(5)$, let $C_{24}^{(5)}$ be the self-dual $[24, 12, 9]$ code of Georgiou and Koukouvinos [7]. Its exact reduction at the ordered coordinate pair $(1, 14)$ gives the self-dual $[22, 11, 8]$ parent $C_{22}^{(5)}$. For the certified parent generator $G_{22}^{(5)}$,

$$x_{22}^{(5)} = (1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 3, 0, 2)$$

has norm -1 , and

$$\text{rowspan } \mathcal{B}_2(G_{22}^{(5)}, x_{22}^{(5)}) = C_{24}^{(5)}.$$

In the same paired coordinate order, the complete normalized generator is

$$\tilde{G}_{24}^{(5)} = \left(\begin{array}{c|c} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 3 & 0 & 2 \\ 1 & 2 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 4 & 4 & 4 & 4 & 4 & 2 & 3 & 1 & 2 & 2 \\ 4 & 3 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 3 & 3 & 3 & 1 & 3 & 4 & 4 & 2 & 4 \\ 1 & 2 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 2 & 4 & 4 & 4 & 3 & 1 & 4 & 2 & 3 \\ 1 & 2 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 4 & 2 & 4 & 3 & 2 & 3 & 4 & 4 & 1 \\ 1 & 2 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 4 & 4 & 2 & 2 & 4 & 2 & 1 & 2 & 1 \\ 1 & 2 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 4 & 2 & 3 & 4 & 2 & 4 & 3 & 1 & 4 & 3 \\ 4 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 & 1 & 2 & 3 & 1 & 3 & 1 & 4 & 1 & 3 & 4 \\ 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 3 & 2 & 4 & 2 & 3 & 4 & 4 & 1 & 1 & 4 & 3 \\ 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 4 & 4 & 4 & 2 & 2 & 2 & 1 & 4 & 4 & 3 \\ 4 & 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 2 & 2 & 3 & 1 & 3 & 1 & 1 & 2 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 4 & 4 & 1 & 1 & 4 & 4 & 4 & 1 & 1 & 4 \end{array} \right).$$

The checker verifies the literal matrix in (21), its parent row space, and the complete weight distribution of $C_{22}^{(5)}$; the exhaustive Magma audit finds 132 distance-nine-loss-free ordered pairs among all $24 \cdot 23 = 552$.

Over $\text{GF}(13)$, let $C_{22}^{(13)}$ be the pure double-circulant self-dual $[22, 11, 10]$ code whose circulant first row is

$$(7, 7, 8, 3, 0, 10, 1, 9, 7, 11, 10),$$

as in [3]. Exact two-coordinate reduction at the ordered pair $(1, 16)$ gives a self-dual $[20, 10, 8]$ code $C_{20}^{(13)}$, and reduction of this same code at $(1, 2)$ gives a self-dual $[18, 9, 7]$ code $C_{18}^{(13)}$. The coordinate indices are one-based, and $C_{22}^{(13)}$ links to the matrices and their verification record.

For the certified parent generators $G_{18}^{(13)}$ and $G_{20}^{(13)}$, the correction vectors are

$$\begin{aligned} x_{18}^{(13)} &= (0, 0, 0, 0, 0, 0, 0, 1, 4, 2, 10, 8, 1, 6, 8, 7, 4, 5), \\ x_{20}^{(13)} &= (0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 7, 7, 8, 3, 10, 1, 9, 7, 11, 10). \end{aligned}$$

They satisfy $x_m^{(13)} \cdot x_m^{(13)} = -1$ and the exact row-space identities

$$\begin{aligned}\text{rowspan } \mathcal{B}_5(G_{18}^{(13)}, x_{18}^{(13)}) &= C_{20}^{(13)}, \\ \text{rowspan } \mathcal{B}_5(G_{20}^{(13)}, x_{20}^{(13)}) &= C_{22}^{(13)}.\end{aligned}$$

The complete normalized generator at the largest level is

$$\tilde{G}_{22}^{(13)} = \left(\begin{array}{cc|cc|cc|cc|cc|cc|cc|cc|cc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 7 & 7 & 8 & 3 & 10 & 1 & 9 & 7 & 11 & 10 \\ 11 & 3 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 9 & 6 & 4 & 2 & 6 & 8 & 9 & 8 & 11 & 4 \\ 12 & 8 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 4 & 3 & 12 & 4 & 6 & 12 & 1 & 7 & 11 & 10 \\ 4 & 7 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 9 & 0 & 3 & 6 & 9 & 7 & 10 & 12 & 6 & 10 \\ 4 & 7 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 11 & 9 & 4 & 9 & 8 & 12 & 0 & 2 & 2 & 2 \\ 2 & 10 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 2 & 10 & 10 & 4 & 1 & 9 & 0 & 4 & 9 & 4 \\ 10 & 11 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 2 & 6 & 11 & 11 & 6 & 4 & 6 & 0 & 9 & 9 \\ 4 & 7 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 2 & 12 & 7 & 8 & 12 & 1 & 4 & 9 & 0 & 4 \\ 7 & 9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 10 & 1 & 9 & 12 & 5 & 8 & 4 & 6 & 0 \\ 8 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & 7 & 12 & 8 & 11 & 2 & 5 & 1 & 4 & 9 \\ 2 & 10 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 8 & 9 & 6 & 6 & 8 & 11 & 12 & 12 & 6 & 1 \end{array} \right).$$

Consequently these are not separately selected examples but one bidirectional lineage

$$C_{18}^{(13)} \longleftrightarrow C_{20}^{(13)} \longleftrightarrow C_{22}^{(13)}.$$

The checker verifies both literal repeated matrices, equality with the corresponding child row spaces, and the complete weight distributions at lengths 18 and 20. The Magma audit examines all $22 \cdot 21 = 462$ ordered pairs at the first reduction and all $20 \cdot 19 = 380$ at the second.

For every upward edge in Table 2, the corresponding checker verifies rank, self-orthogonality, literal reconstruction of the displayed repeated matrix, equality with the child row space, and the recorded weight distribution together with its MacWilliams identity.

Level	Code	Parameters	A_d	Exact upward relation
Binary lineage				
0	C_{16}	[16, 8, 4]	12	base parent
1	C_{18}	[18, 9, 4]	9	$\mathcal{B}_1(G_{16}, x_{16})$
2	C_{20}	[20, 10, 4]	5	$\mathcal{B}_1(G_{18}, x_{18})$
3	C_{22}	[22, 11, 6]	77	$\mathcal{B}_1(G_{20}, x_{20})$
4	$\mathcal{G}_{24}$	[24, 12, 8]	759	$\mathcal{B}_1(G_{22}, x_{22})$
GF(5) lineage				
0	$C_{22}^{(5)}$	[22, 11, 8]	660	base parent
1	$C_{24}^{(5)}$	[24, 12, 9]	1,056	$\mathcal{B}_2(G_{22}^{(5)}, x_{22}^{(5)})$
GF(13) lineage				
0	$C_{18}^{(13)}$	[18, 9, 7]	72	base parent
1	$C_{20}^{(13)}$	[20, 10, 8]	120	$\mathcal{B}_5(G_{18}^{(13)}, x_{18}^{(13)})$
2	$C_{22}^{(13)}$	[22, 11, 10]	23,628	$\mathcal{B}_5(G_{20}^{(13)}, x_{20}^{(13)})$

Table 2: Repeated boxed lineages over GF(2), GF(5), and GF(13).

4.3 A universal rank-one MDS realization over GF(13)

Proposition 4.1 Let $K = \text{GF}(13)$, let $c = 5$, and let

$$G_{14} = [I_7 \mid \text{circ}(2, 9, 10, 9, 2, 1, 1)].$$

This is the pure double-circulant self-dual MDS [14, 7, 8] code recorded in [3]. Put the columns of G_{14} in the order

$$\sigma = (6, 11, 9, 7, 12, 8, 1, 2, 3, 5, 4, 14, 13, 10)$$

and group consecutive coordinates into seven two-coordinate blocks. A row-equivalent universal generator is obtained from the coefficient matrix

$$P = \begin{pmatrix} 10 & 0 & 1 & 12 & 12 & 5 \\ 6 & 5 & 4 & 10 & 11 & 4 \\ 9 & 0 & 4 & 4 & 2 & 3 \\ 0 & 0 & 6 & 9 & 0 & 9 \\ 9 & 0 & 9 & 0 & 9 & 0 \\ 5 & 0 & 1 & 4 & 0 & 9 \end{pmatrix} \quad (23)$$

and

$$H = \begin{pmatrix} 5 \\ 2 \\ 5 \\ 9 \\ 6 \\ 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 3 \\ 9 \\ 9 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad A = (10 \ 4 \ 4 \ 0 \ 0 \ 0) = -Q^\top, \quad D = (1).$$

Let δ_{ij} denote the Kronecker delta: $\delta_{ij} = 1$ when $i = j$, and $\delta_{ij} = 0$ otherwise. It records the identity readout in the second coordinate of the j -th pivot block; explicitly,

$$(P_{ij}, 5P_{ij} + \delta_{ij}) = \begin{cases} (P_{ii}, 5P_{ii} + 1), & i = j, \\ (P_{ij}, 5P_{ij}), & i \neq j. \end{cases}$$

In the notation of Theorem 3.6, this gives

$$\hat{G}_{14} = G(5; P, H, Q, A, D) = \left[\frac{\left((P_{ij}, 5P_{ij} + \delta_{ij}) \right)_{1 \leq i, j \leq 6}}{(A_j(1, 5))_{1 \leq j \leq 6}} \mid \frac{\left((H_i, 5H_i + Q_i) \right)_{1 \leq i \leq 6}}{(1, 5)} \right]. \quad (24)$$

Thus the six pivot rows have exactly the pattern of Theorem 3.6, while every block of the terminal row is a multiple of $(1, 5)$. It has $k = 6$, $r = 1$, and $D = (1)$. Deleting its first block row and first block column gives a self-dual $[12, 6, 6]$ parent. For this displayed parent, the projective correction classes satisfying the necessary weight-six/seven lifting condition form the singleton

$$\Lambda_8 = \{[1 : 8 : 0 : 8 : 3 : 6]\}.$$

The correction column of $\hat{G}_{14}$ is $\gamma = (6, 9, 0, 9, 5, 10)$, whose projective class is the displayed element of Λ_8 ; rebuilding recovers $\hat{G}_{14}$.

Proof. Direct calculation gives $G_{14}G_{14}^T = 0$ and $\text{rank} G_{14} = 7$. Every one of its $\binom{14}{7} = 3432$ seven-column submatrices has rank 7. Hence every seven columns are an information set, so the generated code is MDS and has parameters $[14, 7, 8]$.

For the asserted row equivalence, if G_{14}^σ denotes the stated column reordering, then

$$T = \begin{pmatrix} 12 & 8 & 12 & 5 & 8 & 10 & 0 \\ 10 & 11 & 11 & 4 & 3 & 6 & 0 \\ 4 & 7 & 2 & 3 & 10 & 9 & 0 \\ 9 & 7 & 0 & 9 & 0 & 0 & 0 \\ 0 & 0 & 9 & 0 & 7 & 9 & 0 \\ 4 & 7 & 0 & 9 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 10 & 7 \end{pmatrix}, \quad \det T = 5,$$

and exact multiplication gives $TG_{14}^\sigma = \hat{G}_{14}$, with $\hat{G}_{14}$ as in (24). Applying $\partial(s, t) = t - 5s$ blockwise makes the rank-one structure visible:

$$\partial \hat{G}_{14} = \left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 0 & 0 & 9 \\ 0 & 0 & 1 & 0 & 0 & 0 & 9 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Thus $\text{rank}(\partial\hat{G}_{14}) = 6$, while its last row lies in U_5 and ends in $(1, 5)$; hence $r = 1$ and $D = (1)$. The universal Gram relations hold for the extracted P, H, Q, A, D , so Theorem 3.6 gives self-duality both before and after the stated simultaneous row-and-block deletion.

The parent has $A_j = 0$ for $1 \leq j < 6$, $A_6 = 960$, and $A_7 = 3744$. After projectivizing its weight-six and weight-seven coefficient vectors, there are 392 classes. Exact enumeration of $\mathbf{P}^5(\mathbb{F}_{13})$ shows that precisely $[1 : 8 : 0 : 8 : 3 : 6]$ is nonorthogonal to every one of them. Finally,

$$[(6, 9, 0, 9, 5, 10)] = [1 : 8 : 0 : 8 : 3 : 6],$$

and the lower six rows of (24), after deleting their first block, are literally the parent generator. Reattaching that block and the first row therefore recovers $\hat{G}_{14}$. $\square$

This exact parent–child pair suggests a fixed-seed measure of the construction. Let $C_0 = \langle G_{14} \rangle$, and for $m \geq 0$ define

$$\delta_m(G_{14}) = \max\{d(C_m) : C_{i+1} \text{ is obtained from } C_i \text{ by one non-direct-sum step of Lemma 3.1}\}.$$

The computational question is whether $\delta_m(G_{14})$ reaches the best possible minimum distance of a self-dual code of length $14 + 2m$. The correction-space search first rejects projective correction classes that are orthogonal to a low-weight coefficient line of the fixed parent; exact minimum distance is computed only for the survivors. Any continuation from GF13–14 will be recorded in a separate lineage certificate, so that Table 1 remains a comparison with the published history rather than mixing benchmark and ancestry data.

5 Conclusion

We have studied self-dual codes over finite fields $\mathbb{F}_q$ with $q \equiv 1 \pmod{4}$ from three complementary viewpoints: the classical building-up construction, the hyperbolic geometry forced by the split condition $-1 \in (K^\times)^2$, and a Lean 4 formalization of the resulting algebraic statements. Section 4 exhibits the largest certified repeated boxed presentations over all three fields considered here: the binary Golay endpoint $[24, 12, 8]$, a self-dual $[24, 12, 9]$ code over $\text{GF}(5)$, and a self-dual $[22, 11, 10]$ code over $\text{GF}(13)$. Each is displayed through the same two-coordinate border and an exact certified parent. The universal form also gives an explicit rank-one realization and exact two-coordinate parent of a self-dual MDS $[14, 7, 8]$ code over $\text{GF}(13)$. The algebraic results are formalized in Lean 4.

In the binary case, the universal rank-one normalization proves intrinsically that every Euclidean self-dual code has the literal Chinburg–Zhang boxed representative under a scalar-coordinate permutation. Theorem 3.9 then transports the Chinburg–Zhang cohomological pairing through a specified orthonormal-basis isometry and compares the resulting Euclidean boxed matrix with Kim’s building-up extension. The top-down deletion is a coordinate reduction through a non-alternating first coordinate plane, not deletion of a hyperbolic

pair. Over $\mathbb{F}_q$ with $q \equiv 1 \pmod{4}$, the decomposition obtained from the linear map ∂ in Theorem 3.6 gives the universal analogue: every Euclidean self-dual code has a rank- r boxed representative after a permutation of two-coordinate blocks. The intrinsic integer for the fixed coordinate pairing is $r = \dim(C \cap U_c)$, and the terminal matrix D is nonsingular, with its relation to the other blocks specified by the Gram equations. The explicit split boxed construction is the rank-one specialization with zero pivot diagonal, while the adapted reverse theorem records the relations forced by that normalization. Simultaneous pivot-row and block-column deletion preserves the universal form, giving a chain of self-dual codes ending at $U_c^{(r)}$. Lemma 3.1 gives the converse operation with fixed D , and Lemmas 3.2–3.3 identify its exact fixed-parent relation to Kim–Lee building-up. The case of a zero correction column is a direct-sum extension, not a Kim–Lee step in that fixed coordinate order. These results give a universal representation and recursive construction, not unique parameters or a classification of permutation-equivalence classes. They also do not assert an odd-characteristic arithmetic realization by Hilbert symbols.

The distinction between the bilinear forms is essential. An isometry from the cohomological pairing to the Euclidean form permits comparison with coding-theoretic constructions, whereas a change to hyperbolic coordinates does not imply permutation equivalence. Throughout the normal-form theorems, coordinate permutations preserve the Euclidean form.

Availability of the formalization

The Lean proofs, independent theorem comparisons, and verification instructions, together with the reproducible application data and scripts, are provided in the public repository <https://github.com/LeGenAI/intersection-coding-theory-cohomology>. The version accompanying this manuscript is tagged `afm-revision-2026-08-27`.

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